Discrete Mathematics

Six solved examination papers

Questions preserved from the original scans

Step-by-step solutions for examination preparation

This booklet solves *every* question, including optional questions, in the six supplied papers. Each question is reproduced as an image cut directly from its examination scan. Original wording, notation, numbering, diagrams and printing errors are therefore retained. Read the question image first; the typeset solution follows it. Reproduction of a printed claim does not mean the claim is true: errors and missing assumptions are identified in the solution.

The papers are identified by their *printed dates*, because several filenames do not match their examination headers. The order is: 24 November 2025; 18 November 2022; 23 November 2023; 7 February 2024; 28 January 2023; and 12 November 2019.

Reading the solutions. Every calculation introduces the notation it uses. Algebra proofs verify the required properties rather than merely naming them. In graph drawings, a crossing without a marked vertex is not an additional vertex. An Euler trail uses every edge once; a Hamiltonian path visits every vertex once. A circuit/cycle returns to its starting vertex. An open Euler trail has distinct endpoints.

Recurrences: C.F. + P.I. throughout. The complementary function (C.F.) solves the equation with right-hand side zero. The particular integral (P.I.) solves the given nonzero right-hand side. Their sum is the general solution, after which initial values determine the constants. For homogeneous questions the P.I. is zero. If the usual P.I. trial already occurs in the C.F., multiply the trial by n until it is independent of the C.F. The substitutions are shown in the individual answers.

Compilation. Compile `solutions.tex` twice with XeLaTeX or pdfLaTeX, with the `questions` folder beside it. All question images and the source manifest are included in the source archive. No access to the original file locations is needed.

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1 Final examination: 24 November 2025

Question 1

a)	Let $\mathbb{Z}$ be a set of integers and let R be a relation on $A = \mathbb{Z} \times \mathbb{Z}$ defined by $(a,b)R(c,d)$ *if* $a + d = b + c$, then prove that R is reflexive and transitive.	4

Solution. For two ordered pairs, the defining equation is equivalent to $a - b = c - d$. **Reflexive.** For any (a, b), $a - b = a - b$, so $(a, b)R(a, b)$. **Transitive.** Suppose $(a, b)R(c, d)$ and $(c, d)R(e, f)$. Then $a - b = c - d$ and $c - d = e - f$. Equality gives $a - b = e - f$, or $a + f = b + e$, exactly the condition for $(a, b)R(e, f)$. Thus both required properties hold.

b)	Using the truth table prove that following statement is a Tautology. $[(p \vee q) \wedge (p \longrightarrow r) \wedge (q \longrightarrow r)] \longrightarrow r$	4

Solution. An implication is false only when its antecedent is true and its consequent is false. Let $A = (p \vee q) \wedge (p \to r) \wedge (q \to r)$.

p	q	r	$p \vee q$	$p \to r$	$q \to r$	A	$A \to r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

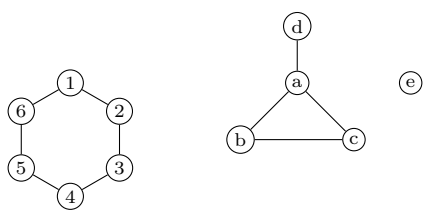
The last column is always true; therefore the statement is a tautology.

c)	In a computer network of 30 computers, each computer is directly connected to at least 10 others. Show that at least two computers have the same number of direct connections.	4

Solution. Treat each computer as a vertex and each direct connection as an undirected edge. A computer cannot connect to itself, so its number of connections is one of $10, 11, \ldots, 29$: only 20 possibilities. Assign the 30 computers to these 20 degree categories. If each category held at most one computer there would be at most 20 computers, a contradiction. Thus at least two have the same degree (in fact at least $\lceil 30/20 \rceil = 2$).

d)	(i) Give an example of a regular, connected graph on six vertices that is not complete. (ii) Identify whether degree sequence (3, 2, 2, 1, 0) is graphic or not. If not then explain your answer, if yes draw graph.	2 2

Solution. (i) Use the cycle C_6, with edges $12, 23, 34, 45, 56, 61$. Each vertex has degree 2, so it is regular; one can follow the cycle between any two vertices, so it is connected. It is not complete because, for example, 1 and 3 are not adjacent. **(ii)** The sequence is graphic. On vertices a, b, c, d, e, use ab, ac, ad, bc. The degrees in that order are $3, 2, 2, 1, 0$.



e)	Let $G = Z_6^* = \{1,2,3,4,5\}$ under multiplication mod 6. Identify all elements that have an inverse. Find the inverse of 5.	4

Solution. An inverse of a modulo 6 is an element b with $ab \equiv 1 \pmod 6$. The products $1 \cdot 1 \equiv 1$ and $5 \cdot 5 = 25 \equiv 1$ show that 1 and 5 are invertible. Multiples of 2 or 3 cannot become congruent to 1 when multiplied by an integer: the product retains that common divisor with 6. Hence $2, 3, 4$ have no inverses. The answer is $\boxed{\{1,5\}, \quad 5^{-1} = 5}$. The displayed five-element set is not itself a group under this operation (for example $2 \cdot 3 \equiv 0$ is outside it).

Question 2

a)	Solve the recurrence relation $a_n = a_{n-1} + a_{n-2}$ with $a_1 = a_2 = 1$	6

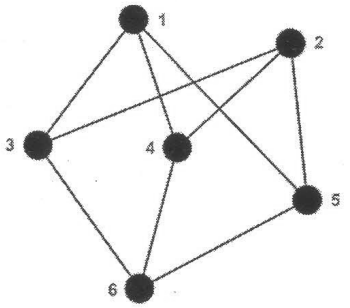
Solution. C.F.. Bring all terms left and try $a_n = r^n$. After division by r^{n-2}, the characteristic equation is $r^2 - r - 1 = 0$. Its roots are $\alpha = (1+\sqrt{5})/2$ and $\beta = (1-\sqrt{5})/2$. Thus $a_n^{(c)} = A\alpha^n + B\beta^n$. **P.I..** The right-hand side is zero, so $a_n^{(p)} = 0$. **Initial values.** The equations are $A\alpha + B\beta = 1$ and $A\alpha^2 + B\beta^2 = 1$. Since $\alpha^2 = \alpha + 1$ and $\beta^2 = \beta + 1$, subtracting the first equation from the second gives $A + B = 0$. Hence $A(\alpha - \beta) = 1$, so $A = 1/\sqrt{5}$, $B = -1/\sqrt{5}$:

$$\boxed{a_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}}.$$

b)	(i) A simple graph has 12 vertices, and each vertex has a degree of either 3 or 5. If the number of vertices with degree 3 is twice the number of vertices with degree 5, determine the number of edges in graph G	6

Solution. Let x count degree-5 vertices. There are $2x$ degree-3 vertices, so $3x = 12$, giving $x = 4$ and $2x = 8$. Each edge contributes 1 to the degree of each endpoint, so $2|E| = \sum \deg v = 4(5) + 8(3) = 44$. Thus $\boxed{|E| = 22}$.

(ii) Check whether the following graph is a bipartite graph. Also write the corresponding adjacency matrix.



Solution. The graph is bipartite with parts $U = \{1, 2, 6\}$ and $V = \{3, 4, 5\}$. Each vertex in U is joined to every vertex in V, and there are no edges within a part: this is $K_{3,3}$. In the requested vertex order $1, 2, 3, 4, 5, 6$, put a 1 precisely where the row and column vertices are adjacent:

$$\boxed{A = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}}.$$

For a bipartite redraw, place $1, 2, 6$ in one column and $3, 4, 5$ in another, and join every pair across columns. The two parts above completely specify that drawing.

c)	If R is the relation defined on $\mathbb{Z}$ (set of integer), such that, $mRn \Leftrightarrow 4$ divides $(m-n)$. Prove that R is an equivalence relation.	8

Solution. To be an equivalence relation, R must be reflexive, symmetric and transitive. For every m, $m-m=0=4\cdot 0$, so mRm. If mRn, write $m-n=4k$ with $k\in\mathbb{Z}$; then $n-m=4(-k)$, so nRm. If also nRp, write $n-p=4\ell$; adding gives $m-p=4(k+\ell)$, so mRp. All three properties hold. Its classes, although not requested, are the four residue classes modulo 4.

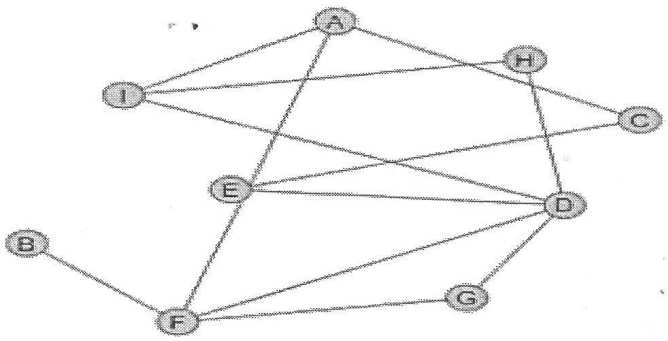
Question 3

a)	Prove by mathematical induction that $n(n+1)(n+2)$ is a multiple of 6 for $n \geq 1$.	6

Solution. Base case. At $n=1$, $1\cdot 2\cdot 3=6$, a multiple of 6. **Induction hypothesis.** Assume $k(k+1)(k+2)=6t$ for some integer t. **Next case.** Subtract the known product from the next one:

$$(k+1)(k+2)(k+3)-k(k+1)(k+2)=3(k+1)(k+2).$$

One of the consecutive integers $k+1, k+2$ is even, so $(k+1)(k+2)=2s$ for an integer s. The next product is therefore $6t+6s=6(t+s)$. This proves the statement for $k+1$, and induction completes the proof.

b)	Determine the chromatic number of the graph given below using Welch Powell's algorithm and write steps of the algorithm. 	6

Solution. Welsh–Powell orders vertices by decreasing degree, gives the first available vertex a colour, and scans the remaining order, adding any vertex not adjacent to a vertex already given that colour. Repeat with a new colour for the remaining vertices. A tie may be settled alphabetically.

Vertex	D	E	F	A	I	C	G	H	B
Degree	5	4	4	3	3	2	2	2	1

The resulting colour classes are $C_1=\{D,A,B\}$, $C_2=\{E,I,G\}$, $C_3=\{F,C,H\}$. No class contains adjacent vertices, giving a 3-colouring. The edges DE, EF, FD form a triangle, whose three mutually adjacent vertices require three different colours. Therefore $\boxed{\chi(G)=3}$; the lower bound proves that the algorithm's result is minimal.

c)	In a survey of 60 people, it was found that; 25 read Business India, 26 reads India Today, 26 reads Times of India, 11 reads both Business India and India Today, 09 reads both Business India and Times of India, 08 reads both India Today and Times of India, 08 reads none of the three. Then find (i) How many read all three? (ii) How many read exactly one?	8

Solution. Let B, I, T denote the three readership sets, and let x be the number reading all three. The union has $60-8=52$ readers. Inclusion–exclusion counts each single set, subtracts the overlaps counted twice, and restores the triple overlap:

$$52=25+26+26-11-9-8+x=49+x, \qquad \boxed{x=3}.$$

The numbers reading only each newspaper are respectively

$$25-11-9+3=8, \quad 26-11-8+3=10, \quad 26-9-8+3=12.$$

Thus $\boxed{\text{exactly one} = 8 + 10 + 12 = 30}$. The pair-only regions are $8, 6, 5$, so $30 + 8 + 6 + 5 + 3 = 52$ checks the union.

Question 4

a)	Prove that the set of non-zero real numbers $\mathbb{R}^*$ forms an abelian group with respect to the operation "$\circ$", where "$\circ$" is defined as $a \circ b = \frac{ab}{2}$, $\forall\, a, b \in \mathbb{R}^*$.	6

Solution. Closure. Nonzero real a, b give nonzero real $ab/2$. **Associativity.** $(a \circ b) \circ c = (ab/2)c/2 = abc/4 = a(bc/2)/2 = a \circ (b \circ c)$. **Identity.** Solving $a \circ e = a$ gives $ae/2 = a$, hence $e = 2$; also $2 \circ a = a$. **Inverse.** The product with an inverse must equal the operation's identity 2. Thus $a \circ b = 2$ means $ab/2 = 2$, so $b = 4/a$, which is nonzero and satisfies both inverse equations. **Commutativity.** $a \circ b = ab/2 = ba/2 = b \circ a$. These verify an abelian group, with $\boxed{e = 2,\ a_{\circ}^{-1} = 4/a}$.

b)	Formalize the following and, by writing truth tables for premises and conclusion, determine whether the following argument is valid. *"If the global warming continues, then the temperature of the world's oceans will rise. If ocean temperatures rise, then hurricanes will become more intense. Therefore, if global warming continues, then hurricane will become more intense."*	6

Solution. Let p mean global warming continues, q mean ocean temperatures rise, and r mean hurricanes become more intense. The premises are $p \to q$, $q \to r$; the conclusion is $p \to r$.

p	q	r	$p \to q$	$q \to r$	$p \to r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

In every row where both premises are true (rows $1, 5, 7, 8$), the conclusion is true. This is precisely the truth-table test of validity. Hence the argument is valid.

c)	Solve the recurrence relation: $a_n - 2a_{n-1} = 3(2^n)$.	8

Solution. C.F.. $a_n - 2a_{n-1} = 0$ has characteristic root 2, so $a_n^{(c)} = C2^n$. **P.I..** A trial $K2^n$ would solve the homogeneous equation and produce zero, so use $a_n^{(p)} = Kn2^n$. Substitution gives

$$Kn2^n - 2K(n-1)2^{n-1} = K2^n = 3 \cdot 2^n.$$

Hence $K = 3$ and $\boxed{a_n = (C + 3n)2^n}$. No initial value is supplied, so the arbitrary constant must remain.

Question 5

a)	Let $\mathbb{R}$ be the set of real numbers and $\mathbb{R}^+$ be the set of positive real numbers. If $f:(\mathbb{R},+)\to(\mathbb{R}^+,\cdot)$ is defined by $$f(x)=e^x, \forall\, x\in\mathbb{R}.$$ Then show that f is a group isomorphism.	6

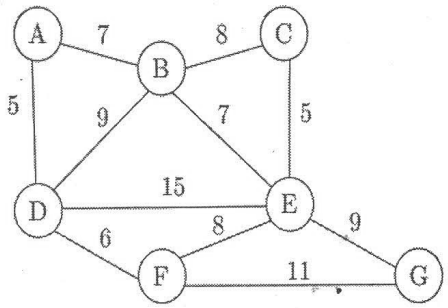
Solution. An isomorphism is a bijection that preserves the operation. Here $f(x+y)=e^{x+y}=e^xe^y=f(x)f(y)$, so addition in the domain becomes multiplication in the codomain. If $f(x)=f(y)$, taking natural logarithms gives $x=y$, proving injectivity. For any target $t>0$, choose $x=\ln t\in\mathbb{R}$; then $f(x)=t$, proving surjectivity. Thus f is a group isomorphism, with inverse ln.

b)	Prove the following statement by contrapositive: For any integers m and n, if the product of m and n is even, then m is even or n is even.	6

Solution. The contrapositive of the printed statement is: if m and n are both odd, then mn is odd. Write $m=2a+1$ and $n=2b+1$, where $a,b\in\mathbb{Z}$. Then

$$mn=4ab+2a+2b+1=2(2ab+a+b)+1,$$

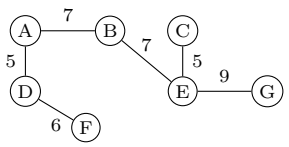
an odd integer. A statement and its contrapositive are logically equivalent, so if mn is even, at least one of m,n must be even.

c)	Construct minimum spanning tree using Kruskal's algorithm. Also find minimum total weight. 	8

Solution. Kruskal's algorithm scans edges from smallest weight upward and accepts an edge only if it does not close a cycle. One accepted sequence is

Edge	AD	CE	DF	AB	BE	EG
Weight	5	5	6	7	7	9

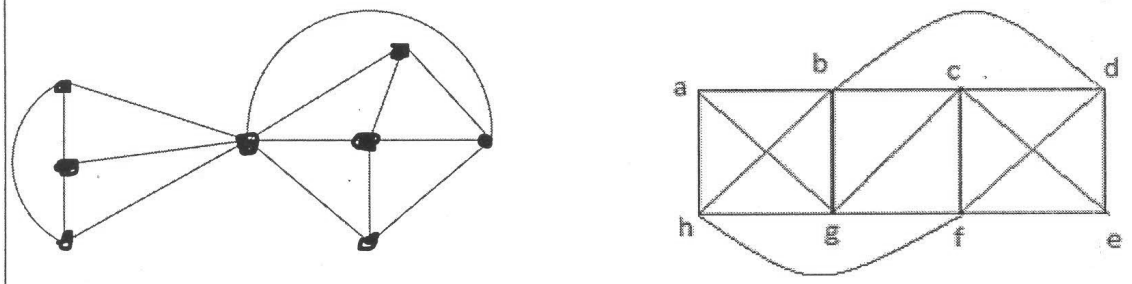
After the first five edges, A,B,C,D,E,F are connected. The weight-8 edges BC, EF would close cycles and are skipped; BD also closes a cycle. The cheapest way to attach the remaining vertex G is EG of weight 9 rather than FG of weight 11. The six accepted edges connect seven vertices without a cycle, so form a spanning tree, with $\boxed{W=5+5+6+7+7+9=39}$.



Question 6

a)	Define conditions for a function to be invertible. Explain why $f(x) = -3x^2 + 7$ is not invertible function from $\mathbb{R} \to \mathbb{R}$.	6

Solution. A function $f : A \to B$ has a two-sided inverse precisely when it is bijective: injectivity gives a unique input for each output, and surjectivity ensures that every element of B has an input. Here $f(1) = f(-1) = 4$ although $1 \neq -1$, so injectivity fails. Also $-3x^2 + 7 \leq 7$ for every real x, so numbers greater than 7 are not attained; surjectivity onto $\mathbb{R}$ fails. Consequently the stated function is not invertible.

b)	Find the edge connectivity and the vertex connectivity of the graphs: 	6

Solution. Write κ for the fewest vertices and λ for the fewest edges whose removal disconnects the graph. **Left graph:** $\boxed{\kappa = 1,\ \lambda = 3}$**.** Denote the common middle vertex by x. Removing x separates the left three vertices from the right four, so $\kappa = 1$. The left three vertices together with x form K_4. On the right, the top, middle and rightmost vertices together with x form another K_4; the bottom, middle and rightmost vertices together with x form a third K_4. Each K_4 remains connected after any two edge deletions: every nontrivial cut of K_4 has at least three edges. Since these cliques share vertices, the whole graph also remains connected. Thus $\lambda \geq 3$. Deleting the three edges incident to a degree-3 vertex isolates it, so $\lambda = 3$.

Right graph: $\boxed{\kappa = \lambda = 3}$**.** The sets $L = \{a, b, g, h\}$ and $R = \{c, d, e, f\}$ each induce a K_4. Across them are bc, gc, gf, bd, hf. After deleting at most two vertices, each surviving clique is connected, and at least one of the three vertex-disjoint cross edges bd, gc, hf survives. Thus two vertices cannot disconnect the graph. Removing b, g, h isolates a, so $\kappa = 3$. Two edge deletions cannot disconnect either K_4 or remove all five cross edges, so $\lambda \geq 3$; isolating degree-3 vertex a gives $\lambda \leq 3$.

c)	For $G = \{0,2,4,6\}$, i) Prepare composition table with respect to addition modulo 8 $(+_8)$. ii) Prove that it is an abelian group with respect to addition modulo 8 $(+_8)$. Is it cyclic? Justify your answer.	8

Solution. Each entry is the row label plus the column label, reduced modulo 8:

$+_8$	0	2	4	6
0	0	2	4	6
2	2	4	6	0
4	4	6	0	2
6	6	0	2	4

All entries lie in G, proving closure. Associativity follows because reducing $(a + b) + c$ or $a + (b + c)$ modulo 8 gives the same residue. The identity is 0, and the inverses of $0, 2, 4, 6$ are $0, 6, 4, 2$. Ordinary addition commutes, so modular addition also commutes. Hence G is abelian. Repeatedly adding 2 gives $2, 4, 6, 0$, so $\boxed{G = \langle 2 \rangle}$ is cyclic (also generated by 6).

Question 7

a)	Using laws of logic prove that $$\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$$	6

Solution. A biconditional is true when both propositions have the same truth value:

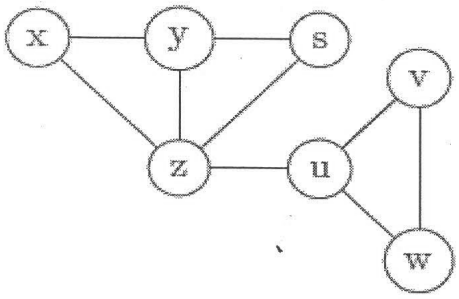
$$\begin{aligned}
\neg(p \leftrightarrow q) &\equiv \neg[(p \wedge q) \vee (\neg p \wedge \neg q)] \\
&\equiv (\neg p \vee \neg q) \wedge (p \vee q) && \text{De Morgan's laws} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge q) \vee (\neg q \wedge p) \vee (\neg q \wedge q) \\
&\equiv (\neg p \wedge q) \vee (p \wedge \neg q). && \text{contradictory terms vanish}
\end{aligned}$$

This is the required expression, in the opposite order of its two disjuncts.

b)	Show that $(Q, +, \times)$ is a ring with unity, where Q is the set of rational numbers.	6

Solution. For rationals a, b, c, sums and products are rational, so both operations are closed. Under addition, $(a+b)+c = a+(b+c)$, $a+b = b+a$, the identity is 0, and the inverse of a is $-a$. Thus $(\mathbb{Q}, +)$ is an abelian group. Multiplication is associative, $(ab)c = a(bc)$, and distributes on both sides: $a(b+c) = ab+ac$, $(a+b)c = ac+bc$. These are identities of ordinary rational arithmetic. Finally $1 \in \mathbb{Q}$ satisfies $1a = a1 = a$. Hence $(\mathbb{Q}, +, \times)$ is a ring with unity $\boxed{1}$ (indeed a commutative ring).

c) (i) Determine whether the given graph has an Eulerian path or an Eulerian circuit. If an Eulerian path or circuit exists, write one such path or circuit explicitly. 8



(ii) Provide an example of a graph that is Eulerian but not Hamiltonian, and an example of a graph that is Hamiltonian but not Eulerian. If such a choice is not possible, explain why.

Solution. (i) The degrees are $\deg x = 2$, $\deg y = 3$, $\deg s = 2$, $\deg z = 4$, $\deg u = 3$, $\deg v = 2$, $\deg w = 2$. The graph is connected and has exactly two odd-degree vertices, y, u. Thus it has an open Euler trail but no Euler circuit. One trail is

$$\boxed{y - x - z - s - y - z - u - v - w - u}.$$

Its nine consecutive edges are distinct and exhaust the graph.

(ii) Two triangles sharing exactly one vertex are Eulerian: every outer vertex has degree 2 and the shared vertex has degree 4. They are not Hamiltonian, since removing the shared vertex disconnects the graph; a Hamilton cycle would leave a connected spanning path after removal of any vertex. Conversely K_4 is Hamiltonian (for instance $1-2-3-4-1$) but is not Eulerian, because all four degrees are odd.

2 Final examination: 18 November 2022

Question 1

Q.1 a.	If A and B are subsets of the universal set U and $n(U) = 50$ $n(A) = 35,\ n(B) = 20,$ and $n(A' \cap B') = 5,$ find i) $n(A \cup B)$ ii) $n(A \cap B)$ iii) $n(A' \cap B)$ iv) $n(A \cap B')$.	[4]

Solution. The complement of $A \cup B$ is $A' \cap B'$, so $n(A \cup B) = 50 - 5 = \boxed{45}$. Since $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, the intersection has $35 + 20 - 45 = \boxed{10}$ elements. Consequently $n(A' \cap B) = 20 - 10 = \boxed{10}$ and $n(A \cap B') = 35 - 10 = \boxed{25}$.

b.	Construct a truth table for $(P \to Q) \wedge (Q \to R)$.	[4]

Solution. Compute each implication first, then take their conjunction (true only if both are true):

P	Q	R	$P \to Q$	$Q \to R$	$(P \to Q) \wedge (Q \to R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

c.	State the generalized pigeon hole principle. Find the minimum number of teachers in a college to be sure that four of them are born in same month.	[4]

Solution. The generalized pigeonhole principle says that assigning N objects to k boxes puts at least $\lceil N/k \rceil$ objects into some box. Here the boxes are the 12 birth months. With 36 teachers, each month could have exactly 3, so four are not guaranteed. With 37, some month has at least $\lceil 37/12 \rceil = 4$. The required minimum is $\boxed{37}$.

d.	Check whether the following graphs exists. Justify your answer. i) A tree with 6 vertices and 5 edges. ii) A graph with degree sequence (2,2,3,3,4)	[4]

Solution. **(i)** Yes. A tree on n vertices has $n - 1$ edges. The path $1 - 2 - 3 - 4 - 5 - 6$ is a connected graph with no cycle and has exactly five edges. **(ii)** Yes. Relabel the sequence in descending order as $(4, 3, 3, 2, 2)$. Use vertices a, b, c, d, e with edges $ab, ac, ad, ae, bc, bd, ce$. Their degrees are respectively $4, 3, 3, 2, 2$, which is the same requested degree multiset.

e.	Show that $\{\mathbb{R}^+, *\}$ where $*$ is defined by $a*b=\frac{ab}{2}$ for all $a,b\in\mathbb{R}^+$ is a monoid where $\mathbb{R}^+$ is the set of all positive integers.	[4]

Solution.

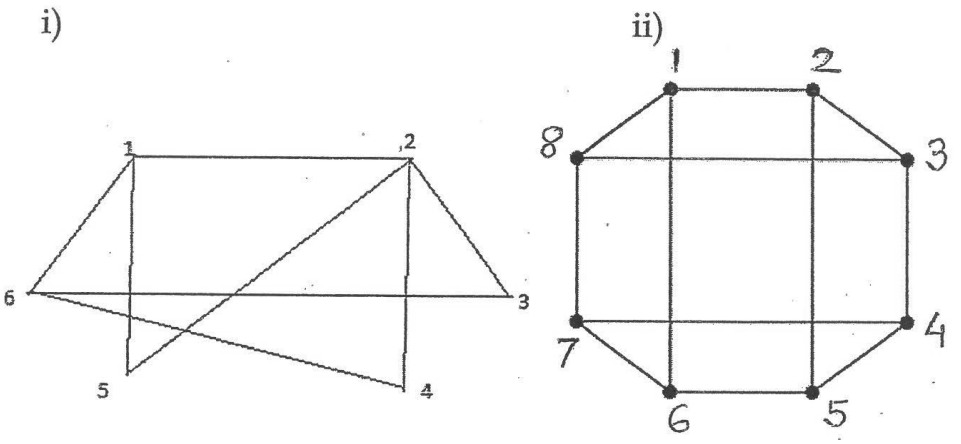
Source note. The scan explicitly calls $\mathbb{R}^+$ "positive integers". Under that wording the claim is false: $1*1 = 1/2$ is not a positive integer, so closure fails. The usual notation $\mathbb{R}^+$ suggests positive real numbers; under that intended reading the following is the proof.

For positive real a, b, $ab/2$ is again positive real. Also $(a * b) * c = abc/4 = a * (b * c)$. The element 2 satisfies $a * 2 = 2 * a = a$. Thus the operation is closed and associative and has an identity, which are precisely the requirements for a monoid.

Question 2

Q.2 a.	Solve the recurrence relation: $a_n = 6a_{n-1} - 9a_{n-2}$, $n \geq 3$, $a_1 = -9, a_2 = -9$	[6]

Solution. C.F.. The homogeneous equation is $a_n - 6a_{n-1} + 9a_{n-2} = 0$, with characteristic polynomial $r^2 - 6r + 9 = (r-3)^2$. A repeated root 3 gives $a_n^{(c)} = (A + Bn)3^n$. **P.I..** It is zero because the equation is homogeneous. **Constants.** $3(A + B) = -9$ and $9(A + 2B) = -9$ give $A + B = -3$ and $A + 2B = -1$. Subtraction gives $B = 2$, then $A = -5$. Therefore $\boxed{a_n = (2n-5)3^n}$. At $n = 1, 2$ this gives $-9, -9$, as required.

b.	Determine which of the following graphs are bi-partite. Justify your answer. 	[6]

Solution. (i) Not bipartite: the edges $12, 25, 51$ form a triangle. Alternating two parts around this odd cycle would force its first and last adjacent vertices into the same part. **(ii)** Bipartite, with parts $\boxed{\{1,3,5,7\},\ \{2,4,6,8\}}$. All eight outer-cycle edges and the chords $16, 25, 38, 47$ run between these two parts. This checks every edge, so the partition proves bipartiteness.

c.	Let $\mathbb{Z}^*$ be a set of non-zero integers and let R be a relation on $A = \mathbb{Z}^* \times \mathbb{Z}^*$ defined by $(a,b)R(c,d)$ *if* $ad = bc$, Then prove that R is an equivalence relation.	[8]

Solution. Reflexive. $ab = ba$, so $(a,b)R(a,b)$. **Symmetric.** If $ad = bc$, then $cb = da$, so $(c,d)R(a,b)$. **Transitive.** Suppose $ad = bc$ and $cf = de$. Multiplying the first by f and substituting the second yields $adf = bcf = bde$, so $d(af - be) = 0$. Since the domain consists of nonzero integers, $d \neq 0$; cancelling it

gives $af = be$, hence $(a,b)R(e,f)$. Therefore R is an equivalence relation. The nonzero condition justifies the cancellation.

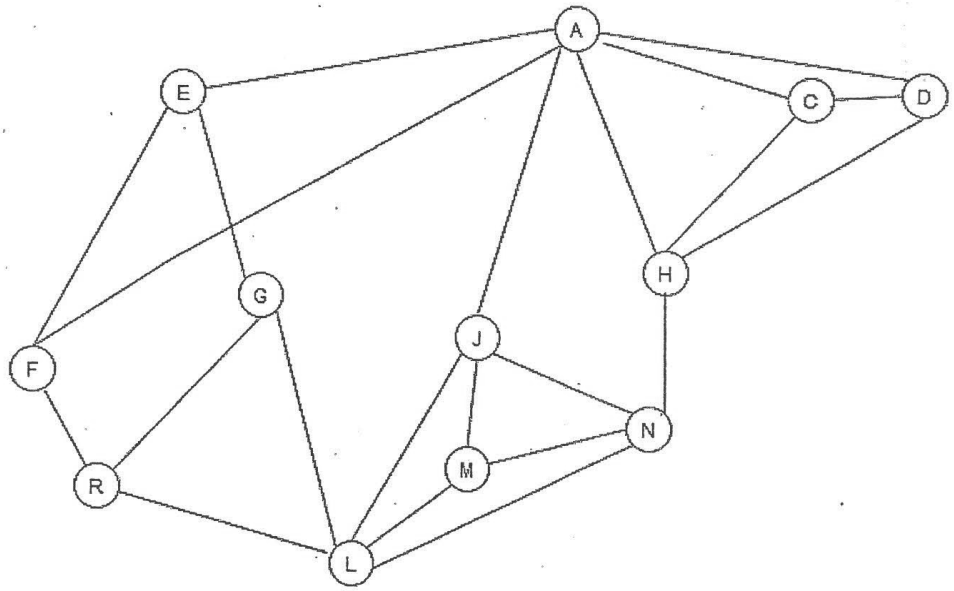
Question 3

Q.3 a.	Using the principle of mathematical induction show that for all positive integers $n \geq 1$, $5^n - 4n - 1$ is divisible by 16.	[6]

Solution. Let $F(n) = 5^n - 4n - 1$. For $n = 1$, $F(1) = 5 - 4 - 1 = 0$, divisible by 16. Assume $F(k) = 16t$. Then

$$F(k+1) = 5^{k+1} - 4k - 5 = 5(5^k - 4k - 1) + 16k = 16(5t + k).$$

Thus the next value is divisible by 16. By induction, this holds for every positive integer n.

b.	Estimate the chromatic number of the following graph using Welch-Powell Algorithm 	[6]

Solution. Using descending degrees (alphabetical tie-breaking) gives:

Vertex	A	L	H	J	N	C	D	E	F	G	M	R
Degree	6	5	4	4	4	3	3	3	3	3	3	3

Scan this order for each colour, adding a vertex only if it is nonadjacent to every vertex already in that colour. Welsh–Powell produces

$$C_1 = \{A, L\}, \quad C_2 = \{H, J, E, R\}, \quad C_3 = \{N, C, F, G\}, \quad C_4 = \{D, M\}.$$

Every listed class is independent, so four colours suffice. The four vertices A, C, D, H are pairwise adjacent: their six edges are AC, AD, AH, CD, CH, DH. Therefore at least four colours are necessary, and $\boxed{\chi(G) = 4}$.

c.	Let f, g be functions from set of real number to set of real number such that $f(x) = 3x - 6$ and $g(x) = \frac{x}{3} + 3$. Find $f \circ g$ and $g \circ f$. Examine whether $f \circ g$ is bijective or not. If $f \circ g$ is bijective find inverse of $f \circ g$.	[8]

Solution. Composition applies the right-hand function first:

$$(f \circ g)(x) = 3(x/3 + 3) - 6 = x + 3, \qquad (g \circ f)(x) = (3x - 6)/3 + 3 = x + 1.$$

The function $h(x) = x+3$ is injective because $h(x) = h(y)$ implies $x+3 = y+3$, hence $x = y$. It is surjective onto $\mathbb{R}$ because, for any real t, the input $x = t-3$ gives $h(x) = t$. Solving $y = x+3$ for x yields $\boxed{(f \circ g)^{-1}(y) = y-3}$.

Question 4

Q.4 a.	Prove that the set of rational numbers $(\mathbb{Q}, \oplus, \otimes)$ is a ring under the operations defined by $a \oplus b = a+b+7 \quad \& \quad a \otimes b = a+b+\frac{ab}{7}$	[6]

Solution. Closure holds because sums, products and division by the nonzero integer 7 preserve rationality. For addition, $(a \oplus b) \oplus c = a+b+c+14 = a \oplus (b \oplus c)$; it commutes, its identity is -7, and the inverse of a is $-14-a$ because $a \oplus (-14-a) = -7$.

For multiplication, direct expansion gives

$$(a \otimes b) \otimes c = a+b+c+\frac{ab+ac+bc}{7}+\frac{abc}{49} = a \otimes (b \otimes c).$$

It commutes. To check distribution rather than assume it, compute

$$a \otimes (b \oplus c) = 2a+b+c+7+\frac{ab+ac}{7} = (a \otimes b) \oplus (a \otimes c).$$

Commutativity gives the other distributive law. Thus the additive operation is an abelian group, multiplication is associative, and both distributive laws hold: this is a commutative ring. It also has multiplicative unity 0, since $a \otimes 0 = a$. The additive zero -7 and multiplicative unity 0 are different elements under these unusual operations.

b.	Convert $\sim(p \to q) \vee ((r \wedge s) \to (q \vee t)) \vee (\sim p \to \sim v)$ into Conjunctive Normal Form.	[6]

Solution. Replace $x \to y$ by $\neg x \vee y$ and use De Morgan's law:

$$\begin{aligned} F &= (p \wedge \neg q) \vee \neg r \vee \neg s \vee q \vee t \vee p \vee \neg v \\ &\equiv p \vee q \vee \neg r \vee \neg s \vee t \vee \neg v. \end{aligned}$$

In the last step, $p \vee (p \wedge \neg q) \equiv p$ (absorption). Thus $\boxed{p \vee q \vee \neg r \vee \neg s \vee t \vee \neg v}$ is CNF: it is a single disjunction of literals, hence a conjunction consisting of one clause. A principal CNF is not requested.

c.	Solve the recurrence relation : $a_n + 4a_{n-1} + 4a_{n-2} = n+1$	[8]

Solution. C.F.. $r^2+4r+4 = (r+2)^2 = 0$, so $a_n^{(c)} = (A+Bn)(-2)^n$. **P.I..** The forcing $n+1$ is linear, and 1 is not a characteristic root, so try $a_n^{(p)} = un+v$. Substituting gives

$$(un+v)+4[u(n-1)+v]+4[u(n-2)+v] = 9un+9v-12u.$$

Match coefficients with $n+1$: $9u = 1$, $9v-12u = 1$. Thus $u = 1/9$, $v = 7/27$.

$$\boxed{a_n = (A+Bn)(-2)^n + \frac{n}{9} + \frac{7}{27}}.$$

No initial values are given, so both constants remain arbitrary.

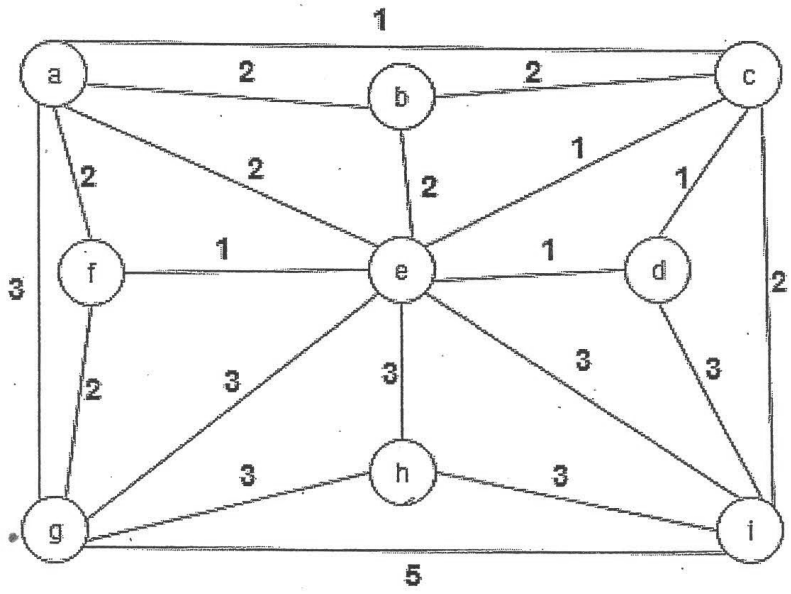
Question 5

Q.5 a.	If G is a group of real numbers under addition and G' is the group of positive real numbers under multiplication, show that the mapping defined by $f(x) = 3^x$ is an isomorphism.	[6]

Solution. The mapping preserves the group operation because $f(x+y) = 3^{x+y} = 3^x 3^y = f(x)f(y)$. It is injective: if $3^x = 3^y$, take logarithms to base 3 to obtain $x = y$. It is onto the positive reals: for each $t > 0$, $x = \log_3 t$ is real and $3^x = t$. Thus it is a bijective homomorphism, hence an isomorphism. Its inverse is $f^{-1}(t) = \log_3 t$.

b.	Prove that $\sqrt{2}$ is an irrational number using method of contradiction.	[6]

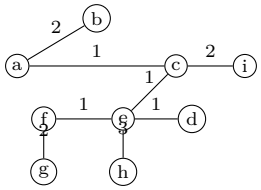
Solution. Assume, for contradiction, that $\sqrt{2} = m/n$ for integers m, n with $n > 0$ and no common divisor greater than 1. Squaring gives $m^2 = 2n^2$, so m^2 is even. The square of an odd integer $(2k+1)^2 = 2(2k^2+2k)+1$ is odd, so m must be even. Write $m = 2k$. Then $4k^2 = 2n^2$, hence $n^2 = 2k^2$, and the same argument makes n even. Both numerator and denominator are divisible by 2, contrary to the fraction being reduced. Therefore $\boxed{\sqrt{2} \text{ is irrational}}$.

c.	Construct a minimum spanning tree using Kruskal's Algorithm. Also find the minimum total weight. 	[8]

Solution. Scan edges in ascending weight, skipping those closing a cycle. One valid accepted sequence is

Edge	ac	ce	ed	ef	ab	fg	ci	eh
Weight	1	1	1	1	2	2	2	3

The weight-1 edge cd is rejected after ce, ed, because it closes triangle $c-e-d-c$. Among weight-2 edges, choose those that attach new vertices b, g, i; all other weight-2 candidates then close cycles. Vertex h has only weight-3 incident edges, and eh attaches it. There are eight accepted edges for nine vertices, all connected and cycle-free. Thus the minimum weight is $\boxed{4(1) + 3(2) + 3 = 13}$.



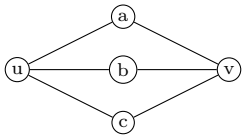
Question 6

Q.6 a.	Let R be a relation on set of real numbers ($\mathbb{R}$), such that aRb if and only if $a^2 + b^2 = 4$. Determine whether the relation R on the set $\mathbb{R}$ is (1) irreflexive (2) antisymmetric (3) transitive with proper justification.	[6]

Solution. Irreflexive: no. Irreflexive means aRa never holds. Here $a = \sqrt{2}$ gives $a^2 + a^2 = 4$, so aRa does hold. (It is not reflexive either, since $0R0$ fails.) **Antisymmetric: no.** Both $0R2$ and $2R0$ hold, yet $0 \neq 2$. **Transitive: no.** $0R2$ and $2R0$ hold, but $0R0$ fails because $0^2 + 0^2 \neq 4$. One counterexample suffices to disprove each universal property.

b.	Identify whether the following graphs are planar or not. Give reason in detail. i) $K_{3,2}$ ii) $K_{3,3}$	[6]

Solution. (i) $K_{3,2}$ is planar. Call the two vertices u, v and the other three a, b, c. Draw three internally disjoint u–v paths $u - a - v$, $u - b - v$, $u - c - v$, above, along and below the middle line:



(ii) $K_{3,3}$ is not planar. In a simple bipartite plane graph every face has boundary length at least 4. Thus $2m \geq 4f$, while Euler's formula gives $n - m + f = 2$, implying $m \leq 2n - 4$. For $K_{3,3}$, $n = 6, m = 9$, but $9 > 2(6) - 4 = 8$. This contradiction rules out a plane drawing.

c.	For $\mathbb{Z}_7 - \{0\}$ i) Prepare composition table with respect to multiplication modulo 7 ii) Prove that it is an abelian group with respect to multiplication modulo 7 iii) Is it cyclic? iv) Find the order of 2 & 4 and subgroups generated by these elements.	[8]

Solution.

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

The table proves closure. Associativity and commutativity descend from ordinary integer multiplication, because reducing a product modulo 7 at an intermediate stage does not change its final residue. The identity is 1. Inverses are $1 \leftrightarrow 1$, $2 \leftrightarrow 4$, $3 \leftrightarrow 5$, $6 \leftrightarrow 6$. Thus it is an abelian group. The powers of 3 modulo 7 are $3, 2, 6, 4, 5, 1$, so it is cyclic. For 2 the powers are $2, 4, 1$; for 4 they are $4, 2, 1$. The first return to the identity is at exponent 3 in each case:

$$\boxed{o(2) = o(4) = 3, \qquad \langle 2\rangle = \langle 4\rangle = \{1, 2, 4\}}.$$

Question 7

Q.7 a.	Apply laws of logic to establish the equivalence: $p \to \big((p \to q) \wedge \sim(\sim q \vee \sim p)\big) \equiv \ \sim p \vee \big(q \wedge p\big)$	[6]

Solution.

$$\begin{aligned} p \to [(p \to q) \wedge \neg(\neg q \vee \neg p)] &\equiv \neg p \vee [(\neg p \vee q) \wedge (q \wedge p)] \\ &\equiv \neg p \vee [(\neg p \wedge p \wedge q) \vee (q \wedge p \wedge q)] \\ &\equiv \neg p \vee (p \wedge q) \equiv \neg p \vee (q \wedge p). \end{aligned}$$

The first line uses implication elimination and De Morgan; the second uses distribution; the third removes a contradiction and repeated literals. This proves the printed equivalence.

b.	Solve the following recurrence relation $a_n + 4a_{n-1} + 4a_{n-2} = 8$ from $n \geq 3$ with $a_1 = 1$, $a_2 = 2$	[6]

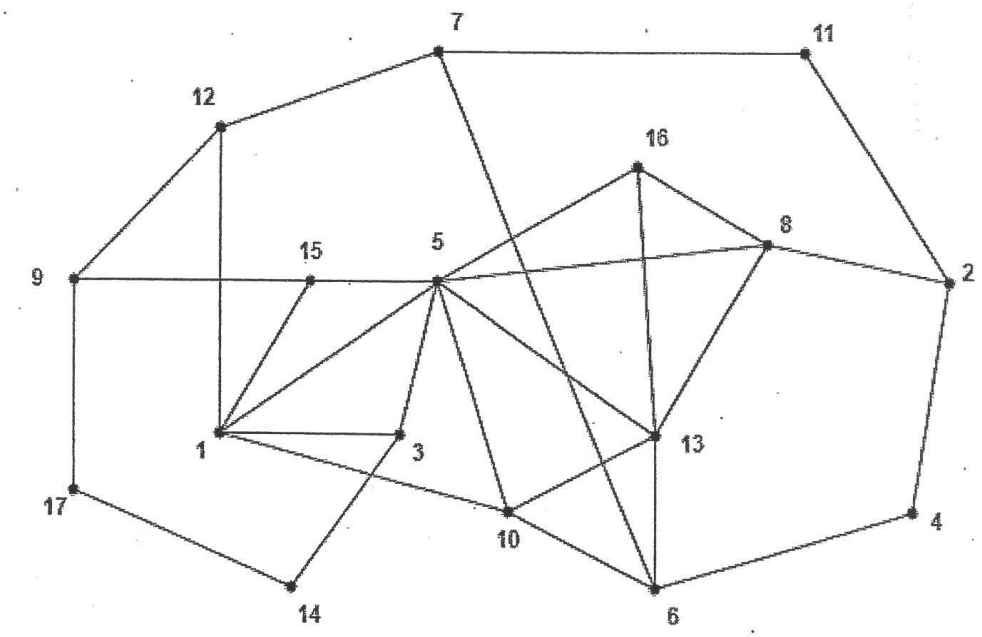
Solution. C.F.. The characteristic polynomial is $(r+2)^2$, so $a_n^{(c)} = (A + Bn)(-2)^n$. **P.I..** Try a constant K. Then $K + 4K + 4K = 8$, giving $K = 8/9$. **Constants.** From $a_1 = 1$ and $a_2 = 2$,

$$-2(A + B) + \frac{8}{9} = 1, \qquad 4(A + 2B) + \frac{8}{9} = 2.$$

Hence $A + B = -1/18$, $A + 2B = 5/18$, so $B = 1/3$, $A = -7/18$. Therefore

$$\boxed{a_n = \left(-\frac{7}{18} + \frac{n}{3}\right)(-2)^n + \frac{8}{9}}.$$

c. Determine whether Eulerian circuit, Hamiltonian path and Hamiltonian cycle exists for the following graph. If so, find them.



[8]

Solution. The degrees, in vertex order $1, 2, \ldots, 17$, are

$$(5, 3, 3, 2, 7, 4, 3, 4, 3, 4, 2, 3, 5, 2, 3, 3, 2).$$

There are ten odd-degree vertices, so an Euler circuit does not exist (nor does an open Euler trail). A Hamiltonian cycle is

$$\boxed{9 - 17 - 14 - 3 - 1 - 12 - 7 - 11 - 2 - 4 - 6 - 10 - 13 - 8 - 16 - 5 - 15 - 9}.$$

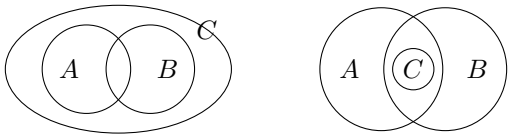
Every consecutive pair is an edge of the given graph, and each of the seventeen vertices occurs exactly once before the return to 9. Deleting its last edge gives the Hamiltonian path $9 - 17 - 14 - 3 - 1 - 12 - 7 - 11 - 2 - 4 - 6 - 10 - 13 - 8 - 16 - 5 - 15$.

3 Final/re-examination: 23 November 2023

Question 1

Q.1 a)	Draw a single Venn diagram to represent the case $A \subseteq C, B \subseteq C$. Hence or otherwise draw another Venn diagram to represent the case $C \subseteq A, C \subseteq B$.	4

Solution. The first case has two overlapping sets A, B inside the larger set C; neither need contain the other. In the second case, C must lie in the intersection of A and B: every element of C lies in both. No inclusion between A and B is required.



The diagrams show one possible configuration in each case; equality of some sets is also allowed by $\subseteq$.

b)	Show that the statement $\big((p \vee q) \wedge (p \to q)\big) \vee \overline{q}$ is a tautology.	4

Solution. Let $F = ((p \vee q) \wedge (p \to q)) \vee \neg q$.

p	q	$p \vee q$	$p \to q$	$(p \vee q) \wedge (p \to q)$	F
T	T	T	T	T	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

Every possible assignment makes F true, so it is a tautology.

c)	20 cards numbered 1 to 20 are placed face down on a table. Cards are selected at random one at a time and turned over until 11 cards have been chosen. If two of the cards add up to 21, the player loses. Is it possible to win this game?	4

Solution. Partition the cards into ten pairs whose sums are 21:

$$\{1, 20\}, \{2, 19\}, \{3, 18\}, \ldots, \{10, 11\}.$$

Choosing eleven distinct cards from ten pairs forces at least two chosen cards into the same pair, by the pigeonhole principle. They add to 21. Therefore the player necessarily loses; winning is impossible. Random choice does not affect this guarantee.

d)	Identify: i) for which values of n are the graphs K_n bipartite? ii) for which values of n are the graphs K_n regular?	4

Solution. (i) K_n is bipartite exactly for $n \leq 2$ (among positive n, $n = 1, 2$). For $n \geq 3$, any three vertices form a triangle, which cannot be bipartite. K_1 and K_2 have immediate two-part partitions, allowing an empty part for K_1. **(ii)** K_n is regular for every positive integer n: every vertex is adjacent to the other $n - 1$ vertices, so every degree is $n - 1$.

e)	Determine whether the set of odd integer together with the operation $a * b = ab$ is a monoid. Justify.	4

Solution. Write two odd integers as $a = 2r + 1$, $b = 2s + 1$. Their product is $ab = 2(2rs + r + s) + 1$, so the set is closed. Multiplication of integers is associative, so $(a * b) * c = (ab)c = a(bc) = a * (b * c)$. The odd integer 1 satisfies $1 * a = a * 1 = a$. Closure, associativity and an identity prove that this is a monoid.

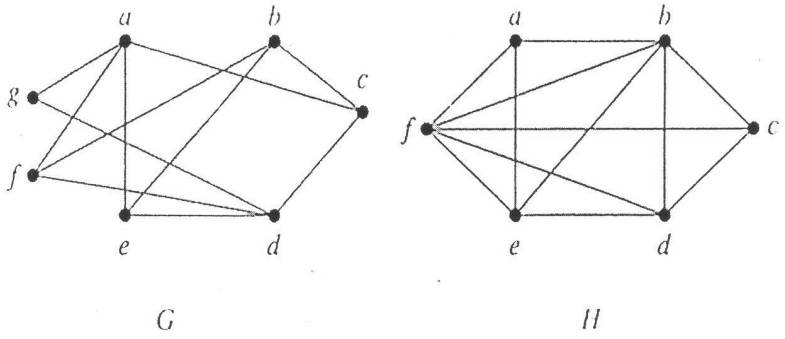
Question 2

Q.2 a)	Determine the solution for recurrence relation $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$ where $a_0 = 0$ and $a_1 = 1$.	6

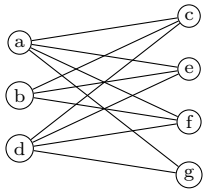
Solution. C.F.. The characteristic equation of $a_n - 2a_{n-1} - a_{n-2} = 0$ is $r^2 - 2r - 1 = 0$, with distinct roots $\alpha = 1 + \sqrt{2}$, $\beta = 1 - \sqrt{2}$. Thus $a_n^{(c)} = A\alpha^n + B\beta^n$. **P.I..** Zero, since the equation is homogeneous. **Constants.** $a_0 = 0$ gives $A + B = 0$. Then $a_1 = 1$ gives $A(\alpha - \beta) = 1$, hence $A = 1/(2\sqrt{2})$, $B = -1/(2\sqrt{2})$.

$$\boxed{a_n = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}}.$$

b)	Are the graphs G and H displayed below bipartite? Justify your answer. If bipartite redraw the graph showing the partitions.	6



Solution. Graph G is bipartite with $U = \{a, b, d\}$ and $V = \{c, e, f, g\}$. Its edges are all cross-part pairs except bg. There are no edges within either part. The following redraw shows the partition:



Graph H is not bipartite: $f - a - b - f$ is an odd cycle of length three. Its edge fb is the sloping line from f to b; crossings along it are not additional vertices.

c)	Let $\mathbb{Z}$ denotes the set of integers. Define a relation R on $\mathbb{Z}$ by setting xRy if and only if $x = y$ or $x = -y$. (i) Examine whether R is equivalence relation or not. (ii) Find all equivalence classes of $\mathbb{Z}$ induced by R.	8

Solution. The condition $x = y$ or $x = -y$ is equivalent to $|x| = |y|$. It is reflexive because $|x| = |x|$; symmetric because equality can be reversed; and transitive because $|x| = |y|$ and $|y| = |z|$ imply $|x| = |z|$. Hence R is an equivalence relation. Its distinct classes are

$$\boxed{[0] = \{0\}, \qquad [k] = \{k, -k\} \quad (k = 1, 2, 3, \ldots)}.$$

There is only one class per positive k: $[k] = [-k]$, so listing both would duplicate it.

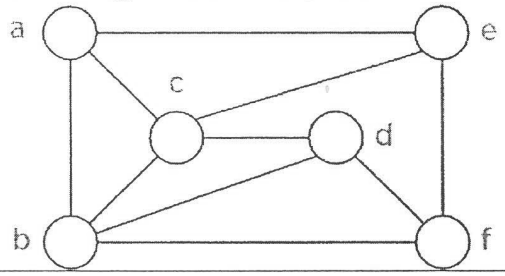
Question 3

Q.3 a)	By mathematical induction, show that $$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$	6

Solution. At $n = 1$, both sides equal 1. Suppose $\sum_{j=1}^{k} j^3 = [k(k+1)/2]^2$. Add $(k+1)^3$ to both sides:

$$\begin{aligned}\sum_{j=1}^{k+1} j^3 &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\ &= \frac{(k+1)^2}{4}[k^2 + 4(k+1)] \\ &= \frac{(k+1)^2(k+2)^2}{4} = \left[\frac{(k+1)(k+2)}{2}\right]^2.\end{aligned}$$

This is the claimed formula with $n = k+1$. Therefore induction proves it for every positive integer n.

b)	Use the Welsh-Powell algorithm to find the chromatic number of the following graph. 	6

Solution. The degrees in descending order are $b : 4, c : 4, a : 3, d : 3, e : 3, f : 3$. Welsh–Powell scans that order and groups nonadjacent vertices as

$$C_1 = \{b, e\}, \qquad C_2 = \{c, f\}, \qquad C_3 = \{a, d\}.$$

For example e is nonadjacent to b, so it may share the first colour; c is adjacent to b, so it must wait for a new colour. All three listed classes are independent. Triangle $a - b - c - a$ requires three colours, so $\boxed{\chi(G) = 3}$.

c)	Let $A = \{1,2,3,4\}$. Examine whether the relation $R = \{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4),(4,3),(4,4)\}$ is symmetric, asymmetric, antisymmetric and transitive. Give argument in support of your answer.	8

Solution. The relation consists of every pair within block $\{1, 2\}$ and every pair within block $\{3, 4\}$. **Symmetric: yes**, since every listed pair has its reverse. **Asymmetric: no**, since $(1, 2)$ and $(2, 1)$ both occur (also loops occur). **Antisymmetric: no**, since those same two pairs occur with $1 \neq 2$. **Transitive: yes**. If xRy and yRz, all three elements belong to the same one of the two blocks. Every pair inside that block is present, so xRz. This verifies every possible composable pair without omitting any case.

Question 4

Q.4 a)	Show that the set of real numbers ($\mathbb{R}$) form an abelian group with respect to binary operation '$*$' defined as $a * b = a + b + 2,\ \forall a, b \in \mathbb{R}$.	6

Solution. Closure follows because $a + b + 2$ is real. Associativity follows from

$$(a * b) * c = (a + b + 2) + c + 2 = a + b + c + 4 = a * (b * c).$$

The equation $a * e = a$ gives $a + e + 2 = a$, so $e = -2$, and $e * a = a$ as well. An inverse b must satisfy $a * b = e = -2$, so $a + b + 2 = -2$ and $b = -a - 4$. This is real and works on both sides. Finally $a * b = a + b + 2 = b + a + 2 = b * a$. Thus it is an abelian group with $\boxed{e = -2,\ a_*^{-1} = -a - 4}$.

b)	Discuss the satisfiability and validity of the following argument: *If you have access to the network, then you can change your grade. You have access to the network. Therefore, you can change your grade.*	6

Solution. Let p mean network access and q mean the ability to change the grade. Premises: $p \to q$ and p. Conclusion: q.

p	q	$p \to q$	$p \wedge (p \to q)$	$[p \wedge (p \to q)] \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

The premises are satisfiable together, because $p = q = \mathrm{T}$ makes both true. The argument is valid, since no row has true premises and a false conclusion; this is modus ponens. Satisfiability asks whether some assignment works, whereas validity asks whether every assignment satisfying the premises also satisfies the conclusion.

c)	Determine the solution of the following recurrence relation: $a_n - 2\,a_{n-1} = 3 \cdot 2^n$, where $n \geq 1$ *and* $a_0 = 1$.	8

Solution. C.F.. The homogeneous equation has root 2, so $a_n^{(c)} = C2^n$. **P.I..** Since 2^n already belongs to the C.F., try $Kn2^n$. Then

$$Kn2^n - 2K(n-1)2^{n-1} = K2^n = 3 \cdot 2^n,$$

giving $K = 3$. **Initial value.** $a_n = (C + 3n)2^n$, and $a_0 = 1$ gives $C = 1$. Thus $\boxed{a_n = (1 + 3n)2^n}$. In particular $a_1 = 8$, and $8 - 2(1) = 6 = 3 \cdot 2$ checks the first step.

Question 5

Q.5 a)	Prove that the additive group of $(\mathbb{Z}_6, +_6)$ is isomorphic to multiplicative group of $(\mathbb{Z}_7^*, \times_7)$. Where $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$.	6

Solution. Use a generator of the multiplicative group: powers of 3 modulo 7 give

k	0	1	2	3	4	5
$f(k) = 3^k \bmod 7$	1	3	2	6	4	5

Define $f([k]_6) = [3^k]_7$. This is well-defined on residue classes because $3^6 \equiv 1 \pmod 7$, so replacing k by $k + 6t$ changes no residue. The six outputs are distinct and exhaust $\mathbb{Z}_7^*$, proving bijectivity. Moreover

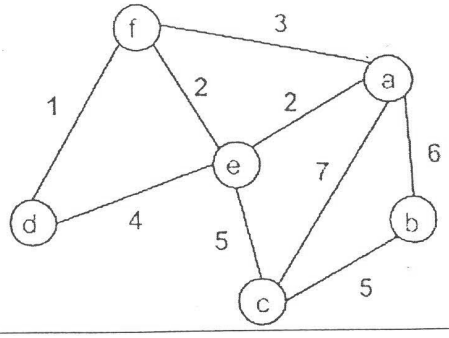
$$f([a]_6 +_6 [b]_6) = [3^{a+b}]_7 = [3^a]_7 \times_7 [3^b]_7 = f([a]_6) \times_7 f([b]_6).$$

Thus f preserves the operations and is an isomorphism.

b)	Prove the following statement with clearly indicate the method of proof used. *If 5n + 3 is odd, then 7n + 4 is even*	6

Solution. Use a direct parity proof, with $n \in \mathbb{Z}$ as understood in this statement. If $5n + 3$ is odd, then $5n = (5n + 3) - 3$ is even, because odd minus odd is even. Since 5 is odd, n must be even: an odd $n = 2k + 1$ would make $5n = 10k + 5$ odd. Write $n = 2k$. Then $7n + 4 = 14k + 4 = 2(7k + 2)$ is even, as required.

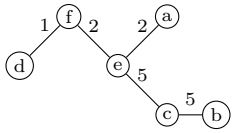
c) Apply Prim's Algorithm to find the minimal spanning tree of the weighted graph: 8



Solution. Prim's algorithm grows one connected tree, each time choosing a smallest-weight edge from a vertex already reached to a vertex not yet reached. Starting at f:

Step	Edge	Weight	New vertex
1	fd	1	d
2	fe	2	e
3	ea	2	a
4	ec	5	c
5	cb	5	b

For example, after f, d, e, a are reached, the eligible cheapest edges to the remaining vertices are ec of weight 5 and ab of weight 6, so choose ec. After reaching c, cb of weight 5 beats ab. These five edges give a tree on all six vertices. The minimum total is $\boxed{1+2+2+5+5=15}$.



Question 6

Q.6 a) Determine whether the function $f\colon \mathbb{R}-\{7\} \longrightarrow \mathbb{R}-\{1\}$ defined as $f(x)=\frac{x+3}{x+7}$ is bijective, if it is then find f^{-1}. 6

Solution.

Source note. As printed, the domain excludes 7, but the denominator $x+7$ vanishes at $x=-7$, which is still in that domain. Consequently the printed formula does not define a function on the stated domain and cannot be bijective there. The natural corrected domain is $\mathbb{R}\setminus\{-7\}$.

For that corrected domain, the value 1 is impossible, since $x+3=x+7$ has no solution. If $f(x)=f(z)$, cross-multiply using the nonzero denominators:

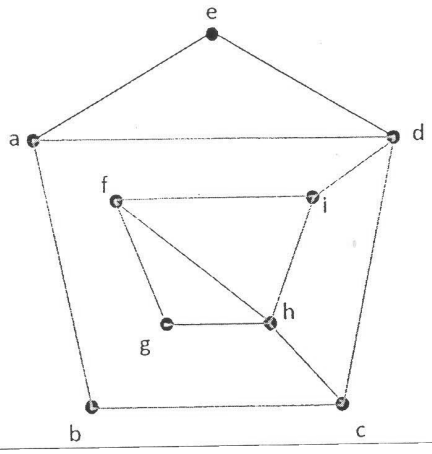
$$(x+3)(z+7)=(z+3)(x+7) \quad\Longrightarrow\quad 4x=4z \quad\Longrightarrow\quad x=z.$$

For any $y\neq 1$, solve $y=(x+3)/(x+7)$:

$$yx+7y=x+3, \qquad x=\frac{3-7y}{y-1}.$$

This input is never -7, since that equation would imply $3=7$. It therefore belongs to the corrected domain and maps to y, proving surjectivity. The corrected bijection has $\boxed{f^{-1}(y)=(3-7y)/(y-1)}$.

b) Find the vertex connectivity and edge connectivity of the following graph. Justify your answer. 6



Solution.

$$\boxed{\kappa(G) = 2, \qquad \lambda(G) = 2}.$$

Removing c, d separates the inner vertices from a, b, e, so $\kappa \leq 2$. No single vertex disconnects the graph: the outer 5-cycle remains connected after one of its vertices is removed; the inner 4-cycle also remains connected after one vertex is removed. The two disjoint connecting edges di, ch ensure at least one outer–inner connection survives a single deletion. Thus $\kappa \geq 2$.

Deleting the two edges di, ch disconnects the inner and outer parts, so $\lambda \leq 2$. No edge is a bridge: every outer or inner edge lies on a cycle, and di, ch both lie on $d - i - h - c - d$. An edge on a cycle has an alternate route between its endpoints, so removing it cannot disconnect the graph. Hence $\lambda \geq 2$.

c) If the addition and multiplication modulo 6 is defined on the set $R = \{0, 1, 2, 3, 4, 5\}$ then test that algebraic structure $(R, +_6, \times_6)$ is a ring with unity. Is it a field? 8

Solution. Addition and multiplication modulo 6 are closed on R. Modular addition is associative and commutative, has identity 0, and the additive inverses of $0, 1, 2, 3, 4, 5$ are $0, 5, 4, 3, 2, 1$. Thus it is an abelian group. Modular multiplication is associative and distributes over modular addition, because the corresponding integer identities remain valid after taking remainders. Its identity is 1. Therefore this is a commutative ring with unity. It is *not* a field: $2 \neq 0$ and $3 \neq 0$, but $2 \times_6 3 = 0$. In particular 2 cannot have a multiplicative inverse, since multiplying $2 \cdot 3 = 0$ by such an inverse would wrongly give $3 = 0$.

Question 7

Q.7 a) Apply laws of logic to establish the equivalence: $p \vee q \vee (\sim p \wedge \sim q \wedge r) \equiv p \vee q \vee r$ 6

Solution. Put $X = p \vee q$. De Morgan gives $\neg p \wedge \neg q = \neg X$. Hence

$$p \vee q \vee (\neg p \wedge \neg q \wedge r) = X \vee (\neg X \wedge r) \equiv (X \vee \neg X) \wedge (X \vee r) \equiv \mathrm{T} \wedge (X \vee r) = p \vee q \vee r.$$

The middle step is the distributive law, and the next uses the law of excluded middle.

b)	Let $G = \{0, 1, 2, 3, 4, 5\}$ set of congruence classes under addition modulo 6 $(+_6)$, (i) Construct composition table with respect to '$+_6$'. (ii) Identify Identity element of $(G, +_6)$. (iii) Identify the inverse of all the elements under $+_6$.	6

Solution. Compute each entry by taking the remainder of the sum on division by 6:

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

The identity is $\boxed{0}$. An additive inverse is an element whose sum is 0, not 1. Thus

$$\boxed{0^{-1} = 0,\ 1^{-1} = 5,\ 2^{-1} = 4,\ 3^{-1} = 3,\ 4^{-1} = 2,\ 5^{-1} = 1}$$

where the superscript denotes inverses for the given addition operation, equivalently $-a \pmod 6$.

c)	Explain Euler Circuit and Hamilton Circuit. Show whether the graph $K_{3,3}$ has Euler Circuit and Hamilton Circuit?	8

Solution. An Euler circuit is a closed trail using every edge exactly once. A Hamilton circuit is a cycle visiting every vertex exactly once before returning to its start. For $K_{3,3}$, name the parts $\{a, b, c\}$ and $\{x, y, z\}$, with all nine cross edges present. Every vertex has degree 3, which is odd, so no Euler circuit exists. A Hamilton circuit does exist:

$$\boxed{a - x - b - y - c - z - a}.$$

All consecutive vertices lie in opposite parts and hence are adjacent; each of the six vertices occurs once before returning to a.

4 Re-examination: 7 February 2024

Question 1

Q.1 a)	Out of 350 candidates who appear in an examination, it was revealed that 150 failed in calculus and 105 in physics. 35 failed in both (calculus and in Physics). Find how many candidates (i) Pass in both (ii) Pass only in calculus.	4

Solution. Let C, P be the sets failing calculus and physics. The number failing at least one is $|C \cup P| = 150 + 105 - 35 = 220$. Thus those passing both are outside this union: $350 - 220 = \boxed{130}$. Passing only calculus means failing physics but not calculus, namely $P \setminus C$, so the number is $105 - 35 = \boxed{70}$. Do not confuse passing only calculus with failing only calculus, whose number would instead be 115.

b)	Show that the statement $(p \vee q) \to (p \wedge q)$ is a contingency.	4

Solution. A contingency is true for some assignments and false for others.

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \to (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

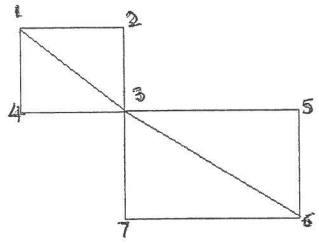
The final column contains both truth values, proving contingency.

c)	Given 12 different 2-digit numbers, show that one can choose two of them so that their difference is a two-digit number with identical first and second digit.	4

Solution. There are only eleven possible remainders modulo 11: $0, 1, \ldots, 10$. Of the twelve distinct two-digit integers, two must have the same remainder. Let the larger be a and the smaller b. Then $a - b$ is a positive multiple of 11. Since $10 \le b < a \le 99$, $0 < a - b \le 89$. Thus

$$\boxed{a - b \in \{11, 22, 33, 44, 55, 66, 77, 88\}},$$

each a two-digit number with identical digits. Distinctness ensures that the difference is not zero.

d)	Does the following graph have an (i) Euler's path (ii) Euler's Circuit? Justify your answer. 	4

Solution. The degrees of vertices 1 through 7 are $(3, 2, 6, 2, 2, 3, 2)$. The connected graph has exactly two odd-degree vertices, $1, 6$. Therefore it has an open Euler path but no Euler circuit. An explicit path is

$$\boxed{1 - 2 - 3 - 4 - 1 - 3 - 5 - 6 - 7 - 3 - 6}.$$

It starts and ends at the two odd vertices and uses each of the ten edges exactly once.

e)	If $\mathbb{Z}_4 = \{0,1,2,3\}$ and $a +_4 b = r$, where r is reminder when we divide $a + b$ by 4. Prove that $(\mathbb{Z}_4, +_4)$ is a group.	4

Solution.

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

The table gives closure and the identity 0. Taking a remainder after either of two additions does not affect the final remainder, so $(a +_4 b) +_4 c$ and $a +_4 (b +_4 c)$ both equal $a + b + c \pmod 4$: associativity holds. The inverses of $0, 1, 2, 3$ are respectively $0, 3, 2, 1$, since each corresponding sum is $0 \pmod 4$. These properties prove it is a group (also abelian, because addition commutes).

Question 2

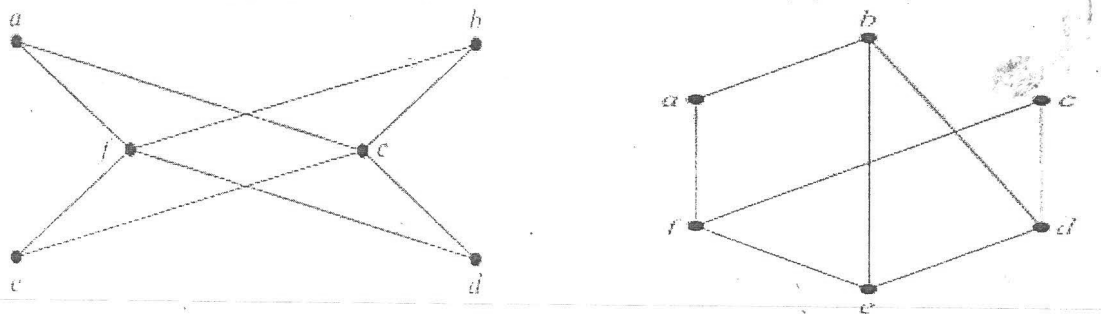
Q.2 a)	Determine the solution for recurrence relation $2a_{n+2} - 11a_{n+1} + 5a_n = 0$ for $n \geq 0$ where $a_o = 2$ and $a_1 = -8$.	6

Solution. C.F.. Substitute $a_n = r^n$ into $2a_{n+2} - 11a_{n+1} + 5a_n = 0$. The characteristic equation factors as

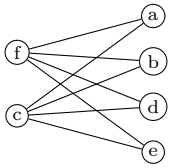
$$2r^2 - 11r + 5 = (2r - 1)(r - 5) = 0.$$

Thus $a_n^{(c)} = A(1/2)^n + B5^n$. **P.I..** Zero, because the forcing is zero. **Constants.** $a_0 = 2$ and $a_1 = -8$ give $A + B = 2$ and $A/2 + 5B = -8$. Substitute $A = 2 - B$ into the second: $1 + (9/2)B = -8$, so $B = -2$, $A = 4$.

$$\boxed{a_n = 4(1/2)^n - 2 \cdot 5^n}.$$

b)	Are the following graphs displayed below bipartite? Justify your answer. 	6

Solution. Left graph: bipartite. The two middle vertices are labelled f, c. Its parts are $U = \{f, c\}$ and $V = \{a, b, d, e\}$: each of f, c is adjacent to all four outer vertices, with no other edges. Hence it is $K_{2,4}$. A partition redraw is



Right graph: not bipartite. Its edges bd, de, eb form a triangle. This odd cycle makes a two-part assignment impossible.

c)	Let $\mathbb{Z}$ denotes the set of integers. Define a relation R on $\mathbb{Z}$ by setting xRy if and only if $x \equiv y \pmod 6$. Examine whether R is equivalence relation or not. Also find the equivalence classes.	8

Solution. Here xRy means $x - y = 6k$ for an integer k. It is reflexive because $x - x = 6 \cdot 0$. If $x - y = 6k$, then $y - x = 6(-k)$, giving symmetry. If $x - y = 6k$ and $y - z = 6\ell$, adding gives $x - z = 6(k + \ell)$, giving transitivity. Thus it is an equivalence relation. The six distinct classes are

$$\boxed{[r] = \{6k + r : k \in \mathbb{Z}\}, \qquad r = 0, 1, 2, 3, 4, 5}.$$

Every integer has exactly one of those remainders; for example $[0] = \{\ldots, -12, -6, 0, 6, 12, \ldots\}$.

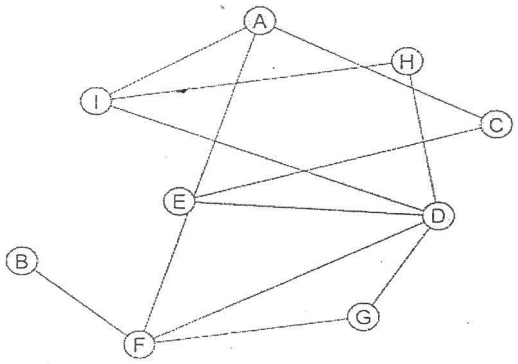
Question 3

Q.3 a)	Show that $2^{2n} - 1$ is divisible by 3 using the principles of mathematical induction.	6

Solution. For $n = 1$, $2^2 - 1 = 3$. Suppose $2^{2k} - 1 = 3t$. Since $2^{2(k+1)} = 4 \cdot 2^{2k}$,

$$2^{2(k+1)} - 1 = 4(2^{2k} - 1) + 3 = 12t + 3 = 3(4t + 1).$$

Thus divisibility holds for $k + 1$ whenever it holds for k. By induction, $3 \mid (2^{2n} - 1)$ for every positive integer n (also for $n = 0$, when the expression is zero).

b)	By using Welsh Powell graph colouring algorithm, find the chromatic number of the following graph. 	6

Solution. The decreasing-degree order is $D(5), E(4), F(4), A(3), I(3), C(2), G(2), H(2), B(1)$. Give the first uncoloured vertex a new colour, then scan the order and add each vertex nonadjacent to every vertex in that colour. This gives

$$C_1 = \{D, A, B\}, \qquad C_2 = \{E, I, G\}, \qquad C_3 = \{F, C, H\}.$$

These independent sets cover every vertex, proving three colours suffice. Triangle $D - E - F - D$ forces at least three. Hence $\boxed{\chi(G) = 3}$.

c)	Let $A = \{1,2,3,4\}$. Examine whether the relation $R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$ is symmetric, asymmetric, antisymmetric and transitive. Give argument in support of your answer.	8

Solution. The listed pairs are precisely the pairs (a, b) with $a < b$ in A. **Symmetric: no**, since $(1, 2)$ occurs but $(2, 1)$ does not. **Asymmetric: yes**, since $a < b$ makes $b < a$ impossible. **Antisymmetric: yes**, since it is impossible to have both aRb and bRa at all; in particular there is no pair of distinct elements violating antisymmetry. **Transitive: yes**, because $a < b$ and $b < c$ imply $a < c$, and the corresponding pair is in the list. Asymmetry and antisymmetry are different properties; a strict-order relation can satisfy both.

Question 4

Q.4 a)	Show that the set of non-zero real numbers ($\mathbb{R}\backslash\{0\}$) form an abelian group with respect to binary operation '$*$' defined as $a * b = (a \times b)/2,\ \forall a, b \in \mathbb{R}\backslash\{0\}$	6

Solution. For nonzero reals a, b, $a * b = ab/2$ is nonzero real, proving closure. Both $(a * b) * c$ and $a * (b * c)$ equal $abc/4$, proving associativity. The identity is 2 because $a * 2 = 2 * a = a$. Solving $a * b = 2$ gives $b = 4/a$, a permitted nonzero real number and a two-sided inverse. Finally $ab/2 = ba/2$, so the operation commutes. All group axioms hold, and the group is abelian, with $\boxed{e = 2,\ a_*^{-1} = 4/a}$.

b)	Discuss the satisfiability and validity of the following argument: *If you study regularly, they you will score distinction. You study regularly. Therefore, you will not score distinction.*	6

Solution. Let p mean regular study and q mean scoring distinction. Premises: $p \to q$ and p; conclusion: $\neg q$. The assignment $p = q = \mathrm{T}$ makes both premises true and the conclusion false. Therefore the argument is *invalid*; the premises in fact force q by modus ponens. The premise set is satisfiable because that same assignment satisfies both premises. If "satisfiability" refers to all premises together with the stated conclusion, their conjunction $(p \to q) \wedge p \wedge \neg q$ is unsatisfiable. This distinction states both possible readings explicitly.

c)	Determine the solution of the following recurrence relation: $a_n - 4\,a_{n-1} = 2 \cdot 4^n$, where $n \geq 1$; *and* $a_0 = 3$.	8

Solution. C.F.. $a_n - 4a_{n-1} = 0$ has solution $C4^n$. **P.I..** Since 4^n already occurs in the C.F., try $Kn4^n$. Substitution gives $Kn4^n - 4K(n-1)4^{n-1} = K4^n = 2 \cdot 4^n$, so $K = 2$. **Initial value.** $a_n = (C + 2n)4^n$, and $a_0 = 3$ gives $C = 3$. Thus $\boxed{a_n = (3 + 2n)4^n}$. At $n = 1$, $a_1 = 20$ and $20 - 4(3) = 8 = 2 \cdot 4$.

Question 5

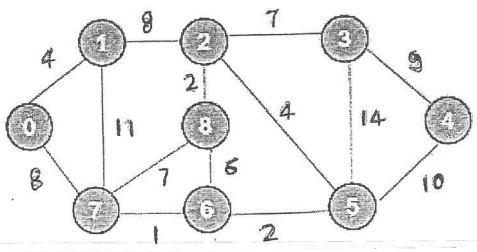
Q.5 a)	Prove that the additive groups $G = (\mathbb{Z}, +)$ and $H = (2\mathbb{Z}, +)$ are isomorphic.	6

Solution. Define $f : \mathbb{Z} \to 2\mathbb{Z}$ by $f(n) = 2n$. This lands in the even integers by definition. It preserves addition since $f(m + n) = 2(m + n) = 2m + 2n = f(m) + f(n)$. It is injective because $2m = 2n$ implies $m = n$. It is surjective because every even integer is $2k$ for some integer k, and $f(k) = 2k$. Thus it is a bijective homomorphism, proving the two additive groups are isomorphic.

b)	Name the methods can be used for proving propositional statements? Prove that there is no smallest positive rational number.	6

Solution. Common proof methods are direct proof, contrapositive, contradiction, proof by cases, and mathematical induction (for statements indexed by integers). A counterexample disproves a universal claim. For the required result use contradiction. Suppose a smallest positive rational r exists. Then $r/2$ is rational, positive, and strictly less than r, since $0 < 1/2 < 1$. This contradicts minimality. Therefore there is no smallest positive rational number.

c) Apply Kruskal's Algorithm to find the minimal spanning tree of the weighted graph: 8

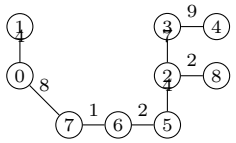


Solution. Kruskal accepts edges in ascending weight whenever their endpoints are not already connected. A valid sequence is

Edge	67	28	56	01	25	23	07	34
Weight	1	2	2	4	4	7	8	9

The weight-6 edge 68 and weight-7 edge 78 would create cycles after the previous choices and are rejected. At weight 8, 07 joins the components containing $0, 1$ and $2, 3, 5, 6, 7, 8$; one could use 12 instead, giving another MST. Vertex 4 is then attached by 34 of weight 9, cheaper than 45 of weight 10. There are eight edges spanning nine vertices, and the minimum total is

$$\boxed{1+2+2+4+4+7+8+9=37}.$$



Question 6

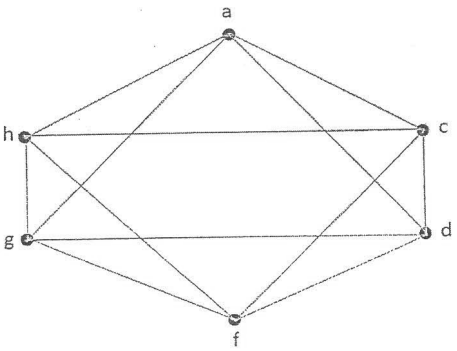
Q.6 a) If $f:\mathbb{R}\longrightarrow\mathbb{R}$, $g:\mathbb{R}\longrightarrow\mathbb{R}$ is defined as $f(x)=7x+x^2$, $g(x)=-7x^3$. Determine the composition functions $g\circ g$, $f\circ g$ and $g\circ f$. 6

Solution. Substitute the inner function's formula in place of the input of the outer function:

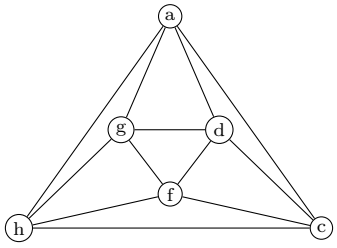
$$(g\circ g)(x) = -7(-7x^3)^3 = 2401x^9,$$
$$(f\circ g)(x) = 7(-7x^3) + (-7x^3)^2 = -49x^3 + 49x^6,$$
$$(g\circ f)(x) = -7(7x+x^2)^3 = -7x^3(x+7)^3.$$

These polynomials are defined for every real input. Leaving the last expression factored is an exact answer.

b) Define the planar graph and determine whether the following graph is Planar. 6



Solution. A planar graph is a graph that admits a drawing in the plane with no crossings except where edges share an endpoint. The given graph *is planar*; crossings in a particular drawing do not prove nonplanarity. It is the octahedral graph. A crossing-free redraw is below:



The redraw retains all twelve original edges: $ac, cd, df, fg, gh, ha, hc, gd, ag, ad, hf, cf$. Since none cross, it certifies planarity.

c)	If the addition and multiplication modulo 7 is defined on the set $R = \{0, 1, 2, 3, 4, 5, 6\}$ then show that algebraic structure $(R, +_7, \times_7)$ is a ring with unity. Is it a field?	8

Solution. Closure, associativity, commutativity and distribution follow by reducing the corresponding integer arithmetic identities modulo 7. Addition has identity 0 and inverses $0, 6, 5, 4, 3, 2, 1$ for elements $0, 1, 2, 3, 4, 5, 6$ respectively, so it is an abelian group. Multiplication has identity 1. Therefore the structure is a commutative ring with unity. It is a field: each nonzero element has a multiplicative inverse. Explicitly $1^{-1} = 1$, $2^{-1} = 4$, $3^{-1} = 5$, $4^{-1} = 2$, $5^{-1} = 3$, $6^{-1} = 6$, as their products are 1 modulo 7. These cover every nonzero element.

Question 7

Q.7 a)	For the given prepositions, identify which are equivalencies or tautologies. i) $p \wedge \sim q$ ii) $\sim p \vee q$ iii) $\sim (\sim p \vee q)$ iv) $\sim p \vee (p \wedge \sim q)$ v) $q \wedge \sim (\sim p \vee q)$	6

Solution. Let the five expressions be $F_1, \ldots, F_5$ in printed order. De Morgan's law gives $F_3 = \neg(\neg p \vee q) = p \wedge \neg q = F_1$. Distribution gives

$$F_4 = \neg p \vee (p \wedge \neg q) = (\neg p \vee p) \wedge (\neg p \vee \neg q) = \neg p \vee \neg q.$$

Finally $F_5 = q \wedge \neg(\neg p \vee q) = q \wedge p \wedge \neg q = \mathrm{F}$.

p	q	F_1	F_2	F_3	F_4	F_5
T	T	F	T	F	F	F
T	F	T	F	T	T	F
F	T	F	T	F	T	F
F	F	F	T	F	T	F

Thus **(i) and (iii) alone are equivalent**. None is a tautology; (v) is a contradiction, and the others are contingencies.

b)	Let $G = \{5, 15, 25, 35\}$ set of congruence classes under multiplication modulo 40 $(\times_{40})$,		6
	(i)	Construct composition table with respect to '$\times_{40}$'.	
	(ii)	Identify Identity element of $(G, \times_{40})$.	
	(iii)	Identify the inverse of all the elements under $\times_{40}$.	

Solution.

$\times_{40}$	5	15	25	35
5	25	35	5	15
15	35	25	15	5
25	5	15	25	35
35	15	5	35	25

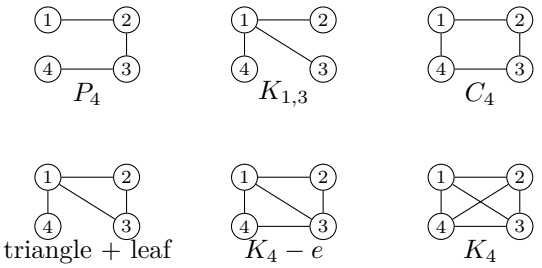
The identity is $\boxed{25}$, since its row and column reproduce the labels. An inverse must multiply to 25, the identity of this restricted structure. The diagonal entries are all 25, so

$$\boxed{5^{-1} = 5, \quad 15^{-1} = 15, \quad 25^{-1} = 25, \quad 35^{-1} = 35}.$$

Using 1 as the identity here would be incorrect: it is not even an element of G.

4; 1; 3	c)	i) Construct all connected graphs of four vertices which are not isomorphic. ii) Construct a graph which is Euler circuit but not Hamiltonian circuit and explain your answer.	8

Solution. (i) Assuming simple undirected graphs, there are exactly six connected nonisomorphic graphs on four vertices:



To see completeness, a connected graph needs at least three edges and has at most six. With three edges it is one of the two four-vertex trees (P_4 or $K_{1,3}$). With four edges its only cycle is either a triangle with an attached leaf or a four-cycle. With five edges it is K_4 minus one edge, and with six it is K_4. The degree sequences distinguish the two graphs with three edges and the two with four edges, so no two in the list are isomorphic.

(ii) Use two triangles $v - a - b - v$ and $v - c - d - v$ sharing only v. Its Euler circuit is $v - a - b - v - c - d - v$. All degrees are even, so this uses every edge. It has no Hamiltonian cycle: removing v leaves two components, whereas removing one vertex of a Hamiltonian cycle would leave a connected path through all remaining vertices.

5 Re-examination: 28 January 2023

Question 1

Q.1 a.	From amongst 2000 literate individuals of a town, 70% read Marathi newspapers, 50% read English newspapers and 32.5% read both Marathi and English newspapers. Find the number of individuals, who read (a) at least one of the newspapers (b) only Marathi newspaper	[4]

Solution. The numbers reading Marathi, English and both are $0.70(2000) = 1400$, $0.50(2000) = 1000$, and $0.325(2000) = 650$ respectively. Inclusion–exclusion gives

$$\boxed{|M \cup E| = 1400 + 1000 - 650 = 1750}.$$

Those reading only Marathi are the Marathi readers outside the overlap: $\boxed{|M \setminus E| = 1400 - 650 = 750}$.

b.	Show that the statement $((p \vee q) \wedge (p \to r) \wedge (q \to r)) \to r$ is a tautology.	[4]

Solution. Let $A = (p \vee q) \wedge (p \to r) \wedge (q \to r)$. The truth table is

p	q	r	$p \vee q$	$p \to r$	$q \to r$	A	$A \to r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

The implication's final column is always true, so it is a tautology.

c.	State pigeon hole principle. An elementary school has 400 students. Show that at least two of them are born on the same day of the year.	[4]

Solution. The pigeonhole principle states that putting more than k objects in k boxes puts at least two objects in one box. Use students as objects and possible birthday dates as boxes. There are at most 366 dates, even allowing 29 February. Since $400 > 366$, at least two students share a birthday date. The result holds whether a 365-day or 366-day calendar is used.

d.	Check whether the following graphs exists. Justify your answer. i) A tree with 6 vertices and 6 edges. ii) A graph with degree sequence (0,1,1,2,3,3)	[4]

Solution. **(i)** Such a tree does not exist. A connected acyclic graph with six vertices has exactly $6 - 1 = 5$ edges, not six. **(ii)** Such a graph exists. On vertices a, b, c, d, e, f, take ab, ac, ad, bc, be. Their degrees are $(3, 3, 2, 1, 1, 0)$, which rearranges to $(0, 1, 1, 2, 3, 3)$ as requested.

e.	Show that $\{\mathbb{Z}^+, *\}$ where $*$ is defined by $a*b=a$ for all $a,b\in\mathbb{Z}^+$ is a semi-group where $\mathbb{Z}^+$ is the set of all positive integers.	[4]

Solution. For any positive integers a, b, $a*b = a$ is again a positive integer, proving closure. To check associativity, $(a*b)*c = a*c = a$, while $a*(b*c) = a*b = a$. Both bracketings give a. Therefore the operation is associative and the structure is a semigroup. A semigroup needs no identity; here no two-sided identity exists, since $e*a = e$ cannot equal every a.

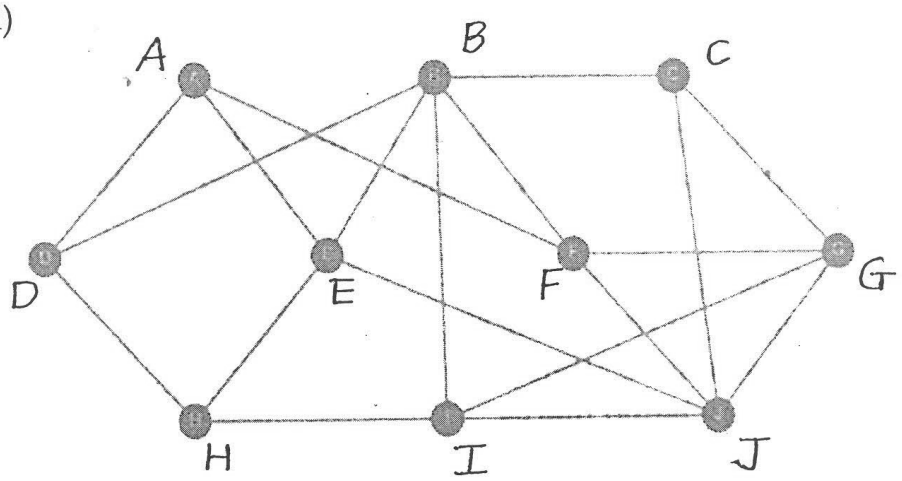
Question 2

Q.2 a.	Determine the Homogeneous solution for the given recurrence relation $a_n - 3a_{n-1} + 2a_{n-2} = 0$ with initial condition $a_0 = 5,\ a_1 = 3$	[6]

Solution. C.F.. The characteristic equation is $r^2 - 3r + 2 = (r-1)(r-2) = 0$, so $a_n^{(c)} = A + B2^n$. **P.I..** Zero because the recurrence is homogeneous. **Constants.** At $n = 0, 1$, the initial values give $A + B = 5$ and $A + 2B = 3$. Subtracting yields $B = -2$, then $A = 7$. Hence $\boxed{a_n = 7 - 2 \cdot 2^n}$. Substitution at $n = 0, 1$ gives $5, 3$.

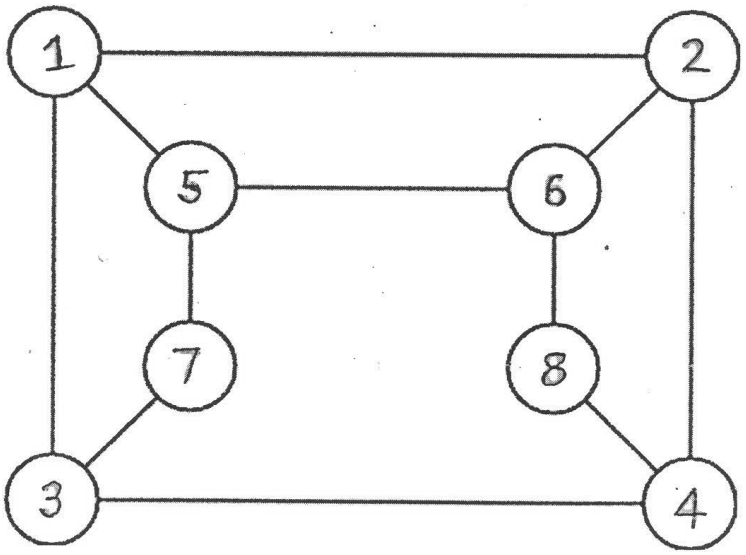
b. Determine which of the following graphs are bi-partite. Justify your answer.

i)



Solution. The first graph is not bipartite. In the displayed labels, F, G, J form triangle $F - G - J - F$. These are the edges across the middle-right, down the right and diagonally from F to J. Every triangle is an odd cycle, so no assignment to two independent parts can exist.

ii)



[6]

Solution. The second graph is bipartite. A valid partition is $\boxed{U = \{1, 4, 6, 7\}, \quad V = \{2, 3, 5, 8\}}$. Its edges are $12, 13, 15, 24, 26, 34, 37, 48, 56, 57, 68$. Each has one endpoint in U and one in V, and the list accounts for all eleven edges. This verifies bipartiteness directly.

c.	Let $\mathbb{Z}$ be a set of integers and let R be a relation on $A = \mathbb{Z} \times \mathbb{Z}$ defined by $(a, b)R(c, d)$ *if* $a + d = b + c$. Then prove that R is an equivalence relation.	[8]

Solution. Rewrite the defining equation $a + d = b + c$ as $a - b = c - d$. Reflexivity holds because $a - b = a - b$. Symmetry holds because $a - b = c - d$ implies $c - d = a - b$, equivalent to $(c, d)R(a, b)$. For transitivity, if $(a, b)R(c, d)$ and $(c, d)R(e, f)$, then $a - b = c - d = e - f$, hence $a + f = b + e$, meaning $(a, b)R(e, f)$. All three equivalence-relation properties are proved.

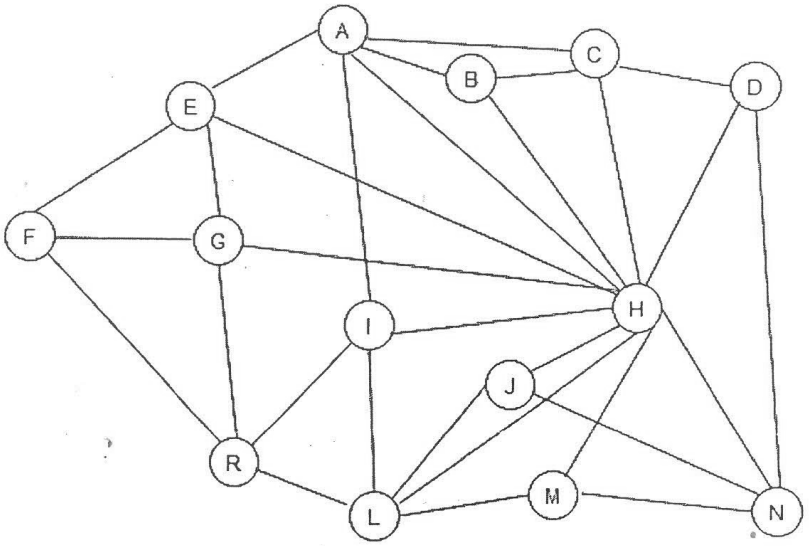
Question 3

Q.3 a.	Prove that $1 + 2 + 2^2 + 2^3 + \ldots\ldots\ldots + 2^n = 2^{n+1} - 1$ for all integers $n \geq 1$, using the principle mathematical induction.	[6]

Solution. Interpret the printed string as $1 + 2 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 1$ (the last superscripts are poorly printed). At $n = 1$, $1 + 2 = 3 = 2^2 - 1$. Assume $1 + 2 + \cdots + 2^k = 2^{k+1} - 1$. Adding the next term 2^{k+1} gives

$$1 + 2 + \cdots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1.$$

This is the formula at $k + 1$, completing induction.

b.	Estimate the chromatic number of the following graph using Welch-Powell Algorithm 	[6]

Solution. A decreasing-degree order, breaking ties alphabetically, is

Vertex	H	A	L	C	E	G	I	N	R	B	D	F	J	M
Degree	11	5	5	4	4	4	4	4	4	3	3	3	3	3

Welsh–Powell scans that order for each colour, giving

$$C_1 = \{H, R\}, \quad C_2 = \{A, L, G, N\}, \quad C_3 = \{C, E, I, J, M\}, \quad C_4 = \{B, D, F\}.$$

Vertices within each class are pairwise nonadjacent. Thus four colours suffice. Vertices A, B, C, H have all six mutual edges AB, AC, AH, BC, BH, CH, so form K_4 and require four colours. Therefore $\boxed{\chi(G) = 4}$.

c.	Let f, g be functions from set of real number to set of real number such that $f(x) = 2x + 5$ and $g(x) = \frac{x}{3} - 7$. Find $f \circ g$ and $g \circ f$. Examine whether $f \circ g$ is bijective or not. If $f \circ g$ is bijective find inverse of $f \circ g$.	[8]

Solution.

$$(f \circ g)(x) = 2(x/3 - 7) + 5 = \frac{2x}{3} - 9, \quad (g \circ f)(x) = \frac{2x + 5}{3} - 7 = \frac{2x}{3} - \frac{16}{3}.$$

For $h = f \circ g$, equality $h(x) = h(z)$ implies $2x/3 - 9 = 2z/3 - 9$, hence $x = z$, proving injectivity. For any real y, the equation $y = 2x/3 - 9$ has the real solution $x = 3(y + 9)/2$, proving surjectivity. Therefore h is bijective and $\boxed{h^{-1}(y) = \frac{3}{2}(y + 9)}$.

Question 4

Q.4 a.	Prove that the set of real numbers $(\mathbb{R}, \oplus, \otimes)$ is a ring under the operations defined by $a \oplus b = a + b - 1$ & $a \otimes b = a + b - ab$	[6]

Solution. Both formulas give real outputs for real inputs, so closure holds. For addition,

$$(a \oplus b) \oplus c = a + b + c - 2 = a \oplus (b \oplus c),$$

and it commutes. Its identity is 1, since $a \oplus 1 = a$, and its inverse of a is $2 - a$, since $a \oplus (2 - a) = 1$. Thus $(\mathbb{R}, \oplus)$ is an abelian group. For multiplication, expanding either bracketing gives

$$(a \otimes b) \otimes c = a + b + c - ab - ac - bc + abc = a \otimes (b \otimes c).$$

It also commutes. Finally

$$a \otimes (b \oplus c) = 2a + b + c - 1 - ab - ac = (a \otimes b) \oplus (a \otimes c).$$

This is one distributive law, and commutativity gives the other. Hence the structure is a commutative ring; it even has multiplicative unity 0 because $a \otimes 0 = a$. The additive zero is 1, not the ordinary real zero.

b.	Convert $q \vee (p \wedge r) \wedge \sim((p \vee r) \wedge q)$ into Disjunctive Normal Form.	[6]

Solution. Use the standard precedence convention: $\neg$ first, then $\wedge$, then $\vee$. The expression is $q \vee [(p \wedge r) \wedge \neg((p \vee r) \wedge q)]$. By De Morgan and distribution,

$$\begin{aligned} F &\equiv q \vee [(p \wedge r) \wedge ((\neg p \wedge \neg r) \vee \neg q)] \\ &\equiv q \vee (p \wedge r \wedge \neg p \wedge \neg r) \vee (p \wedge r \wedge \neg q) \\ &\equiv q \vee (p \wedge r \wedge \neg q) \\ &\equiv (q \vee \neg q) \wedge [q \vee (p \wedge r)] \equiv \boxed{q \vee (p \wedge r)}. \end{aligned}$$

This is disjunctive normal form: an OR of conjunctions of literals (a single literal q is an allowed term).

c.	Evaluate the total solution of the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 6(5^n)$	[8]

Solution. C.F.. The characteristic polynomial is $r^2 - 5r + 6 = (r - 2)(r - 3)$, so $a_n^{(c)} = A2^n + B3^n$. **P.I..** Since 5 is not a characteristic root, try $a_n^{(p)} = K5^n$. Its substitution is

$$K5^{n+2} - 5K5^{n+1} + 6K5^n = K5^n(25 - 25 + 6) = 6K5^n.$$

Matching $6 \cdot 5^n$ gives $K = 1$. Thus the total solution is $\boxed{a_n = A2^n + B3^n + 5^n}$. No initial data are provided, so A, B cannot be fixed.

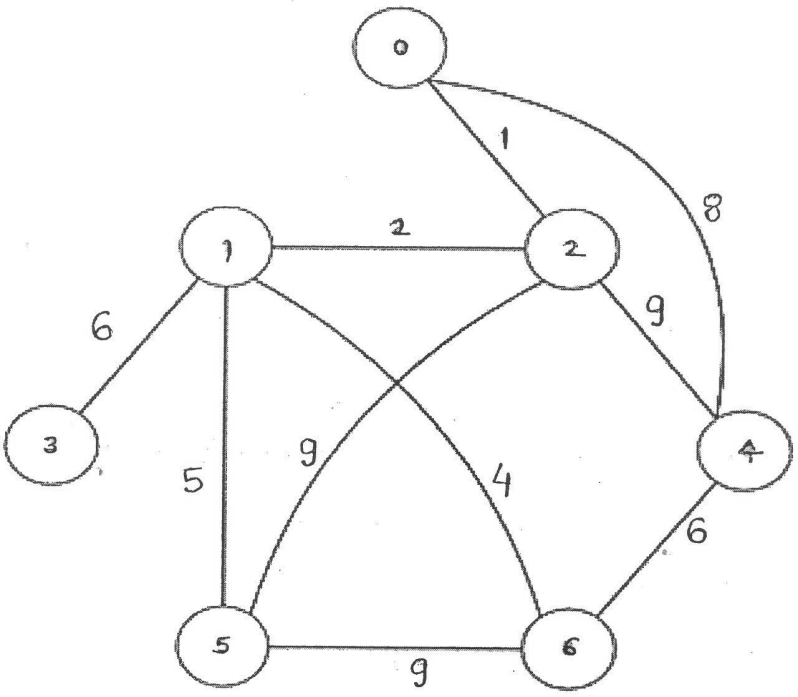
Question 5

Q.5 a.	G is a group of non-zero real numbers under multiplication and n is a positive integer. Check whether $f(x) = x^n$ is an isomorphism from G to G.	[6]

Solution. For every positive integer n, f preserves multiplication: $f(xy) = (xy)^n = x^n y^n = f(x)f(y)$. Whether it is an isomorphism depends on parity. **If n is odd**, each nonzero real t has a unique real nth root x, which is also nonzero. This proves existence and uniqueness of the preimage, hence bijectivity. Equivalently an odd power is strictly increasing on $\mathbb{R}$. Thus f is an isomorphism with inverse the real nth-root function. **If n is even**, $f(1) = f(-1) = 1$, so it is not injective. Its outputs are all positive, so negative target elements are missed, and it is not onto G. Therefore $\boxed{f \text{ is an isomorphism exactly when } n \text{ is odd}}$.

b.	Prove that $\sqrt{3}$ is an irrational number using method of contradiction.	[6]

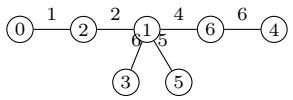
Solution. Assume $\sqrt{3} = m/n$ in lowest terms, with integers m, n, $n > 0$. Squaring gives $m^2 = 3n^2$. Any integer is $0, 1$ or 2 modulo 3, and its square is respectively $0, 1$ or 1 modulo 3. Thus $3 \mid m^2$ forces $3 \mid m$. Write $m = 3k$. Then $9k^2 = 3n^2$, so $n^2 = 3k^2$, and the same square-remainder argument gives $3 \mid n$. This contradicts the fraction being in lowest terms. Hence $\boxed{\sqrt{3} \text{ is irrational}}$.

c.	Construct a minimum spanning tree using Prims Algorithm. Also find minimum total weight. 	[8]

Solution. Start Prim's algorithm at 0. At each step choose a minimum-weight edge crossing from the reached vertices to an unreached vertex:

Step	Edge	Weight	New vertex
1	02	1	2
2	21	2	1
3	16	4	6
4	15	5	5
5	13	6	3
6	64	6	4

After the first five choices, the only unreached vertex is 4. The crossing candidates to it are 64 of weight 6, 04 of weight 8, and 24 of weight 9, so choose 64. All seven vertices are reached through six accepted edges. The MST weight is $\boxed{1+2+4+5+6+6=24}$.

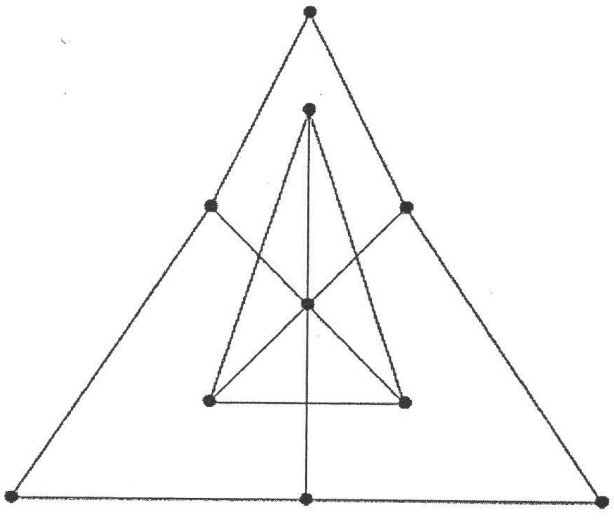


Question 6

Q.6 a. Let R be a relation on set of integers ($\mathbb{Z}$), such that aRb if and only if $a \leq b + 1$. Determine whether the relation R on the set $\mathbb{Z}$ is (1) irreflexive (2) antisymmetric (3) transitive with proper justification. **[6]**

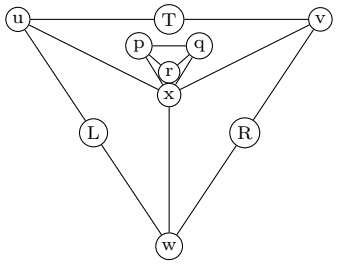
Solution. Irreflexive: no. For every integer a, $a \leq a + 1$, so every aRa holds; the relation is actually reflexive. **Antisymmetric: no**. $0R1$ because $0 \leq 2$ and $1R0$ because $1 \leq 1$, but $0 \neq 1$. **Transitive: no**. $2R1$ and $1R0$ hold, since $2 \leq 2$ and $1 \leq 1$. However $2R0$ would require $2 \leq 1$, which is false.

b. Identify whether the following graph is planar or not. Give reason in detail. **[6]**



Solution. The graph is planar. To make the unlabelled drawing precise, name its outer corners T (top), L (bottom left), R (bottom right), and its outer side midpoints u (left), v (right), w (bottom). Call the central marked vertex x, and the inner triangle's vertices p (top), q (left), r (right). The graph comprises the outer subdivided triangle $u - T - v - R - w - L - u$, spokes xu, xv, xw, and the inner K_4 on x, p, q, r.

The outer part is a planar K_4 with three edges subdivided; the inner part is another planar K_4 sharing only x. It can be placed entirely within a face incident to x. Here is a crossing-free redraw retaining every edge:



Thus the crossings in the original layout can be removed; they are not evidence of nonplanarity.

c.	Let G= {0, 1, 2, 3, 4, 5} be a set of congruence classes under addition modulo 6 i) Prepare composition table with respect to addition modulo 6 ii) Prove that G is an abelian group with respect to addition modulo 6 iii) Find the inverse of elements 2, 3 and 5. iv) Is it cyclic? v) Find the order of elements 2, 3 and sub groups generated by these elements.	[8]

Solution.

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

The table proves closure and identity 0. Associativity and commutativity follow from integer addition after reduction modulo 6. Every a has inverse $-a \pmod 6$, so this is an abelian group. In particular, under addition,

$$\boxed{2^{-1} = 4, \qquad 3^{-1} = 3, \qquad 5^{-1} = 1}.$$

It is cyclic because repeated addition of 1 gives $1, 2, 3, 4, 5, 0$, covering all elements. For 2, the successive multiples are $2, 4, 0$, so $\boxed{o(2) = 3,\ \langle 2\rangle = \{0, 2, 4\}}$. For 3, they are $3, 0$, so $\boxed{o(3) = 2,\ \langle 3\rangle = \{0, 3\}}$.

Question 7

Q.7 a.	Apply laws of logic to establish the equivalence: $\sim(p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q)) \equiv (\sim p \vee q)$	[6]

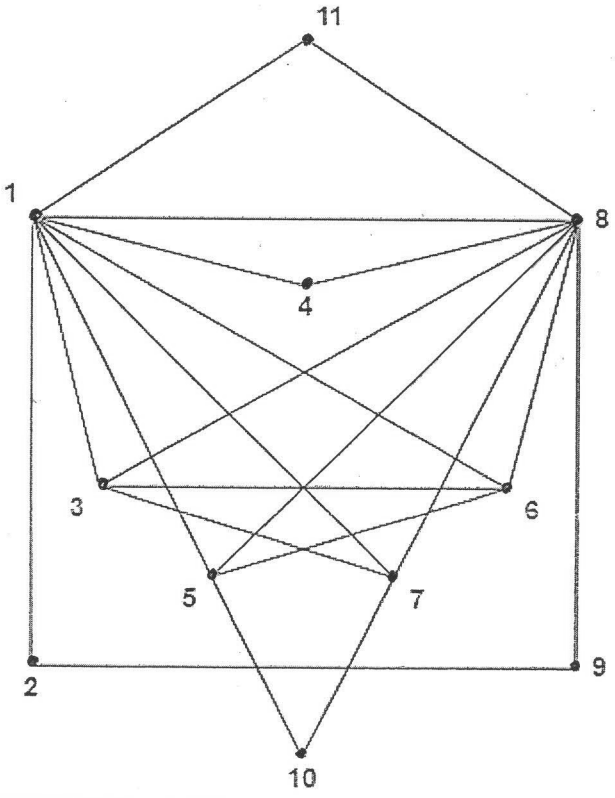
Solution.

$$\begin{aligned}
\neg(p \wedge q) \rightarrow [\neg p \vee (\neg p \vee q)] &\equiv \neg\neg(p \wedge q) \vee \neg p \vee q && \text{implication; idempotence} \\
&\equiv (p \wedge q) \vee \neg p \vee q && \text{double negation} \\
&\equiv \neg p \vee q. && \text{absorption: } q \vee (p \wedge q) = q
\end{aligned}$$

This is exactly the printed right-hand side.

b.	Solve the given recurrence relation to get the total solution: $a_n - a_{n-1} - 6a_{n-2} = 4$	[6]

Solution. C.F.. $r^2 - r - 6 = (r-3)(r+2) = 0$ gives $a_n^{(c)} = A3^n + B(-2)^n$. **P.I..** Try a constant K, since the right-hand side is constant and 1 is not a characteristic root. Substitution gives $K - K - 6K = 4$, so $K = -2/3$. Thus $\boxed{a_n = A3^n + B(-2)^n - \frac{2}{3}}$. No initial values are supplied, so both constants remain arbitrary.

c.	Determine whether Eulerian trail, Eulerian circuit, Hamiltonian path exists for the following graph. If so, find them.  	[8]

Solution. The degrees in the order $1, 2, \ldots, 11$ are $(8, 2, 4, 2, 4, 4, 4, 8, 2, 2, 2)$. The graph is connected and all degrees are even, so it has an Euler circuit. One is

$$\boxed{1-2-9-8-1-3-8-4-1-5-8-6-1-7-10-5-6-3-7-8-11-1}.$$

There are 21 successive edges, all distinct, matching half the degree sum $42/2 = 21$. This circuit is also an Euler trail in the general sense. If "trail" is intended to mean an *open* Euler trail, none exists, since that would require exactly two odd-degree vertices.

No Hamiltonian path exists. Remove vertices $1, 8$. The remaining graph has four connected components:

$$\{11\}, \quad \{4\}, \quad \{2, 9\}, \quad \{3, 5, 6, 7, 10\}.$$

Deleting two vertices from any path leaves at most three path pieces. If a Hamiltonian path had existed, those pieces would cover all remaining vertices and force at most three components, contradicting the four just listed. This proves nonexistence rather than merely failing to find a path.

6 Final examination: 12 November 2019

Question 1

Q. 1) A) **a)** If $A \subseteq B$ and $B \subseteq C$ then $A \cup (B \cap C)$ equals to _____ [1]

b) If the universe of discourse is {1,2}, then the truth value of $(\exists x)(x^5 + x^4 = 12)$ [1] equals to _____

c) If $A = \{1,2,3\}$ then total number of equivalence relations on A are ____ [1]

d) Letters of the word ASSASSINATION can be arranged in _____ number of ways so [1] that all the 'S's are together.

i) 151200 ii) 151212 iii) 151324 iv) 151222

e) If $a_0 = 3$ and $a_{n+1} = 2a_n + 3$ then a_1, a_2, a_3, a_4 are given by [1]

i)$a_1 = 7, a_2 = 23, a_3 = 45, a_4 = 93$

ii)$a_1 = 8, a_2 = 22, a_3 = 44, a_4 = 93$

iii)$a_1 = 9, a_2 = 21, a_3 = 46, a_4 = 94$

iv)$a_1 = 9, a_2 = 21, a_3 = 45, a_4 = 93$

f) $(2Z, .)$ is _____ , where ' . ' represents usual multiplication [1]

i) Groupoid ii) Semi-Group iii) Monoid iv) Group

g) Permutation group S_3 contains _____ number of elements of order 3. [1]

h) Which of the following isbipartite? [1]

i) $K_n, n \geq 3$ ii)$K_{m,n}$ iii) C_n, n is even iv) both ii and iii

i) Chromatic number ' χ 'of tree is _____ [1]

j) The statement $p \rightarrow (q \rightarrow p)$ is _____ [1]

i) contingency ii) contradiction iii) tautology

Solution. A(a) Since $B \subseteq C$, $B \cap C = B$. Since $A \subseteq B$, $A \cup B = B$. Answer: $\boxed{B}$.

A(b) Test the two permitted inputs: $1^5 + 1^4 = 2$ and $2^5 + 2^4 = 48$. Neither is 12. The existential statement is therefore $\boxed{\text{False}}$.

A(c) Equivalence relations correspond to partitions. The five partitions of $\{1, 2, 3\}$ are $\{123\}$, $\{1|23\}$, $\{2|13\}$, $\{3|12\}$, $\{1|2|3\}$, where a bar separates blocks. Each relation connects exactly the elements in the same block. Answer: $\boxed{5}$.

A(d) In ASSASSINATION there are 3 A's, 4 S's, 2 I's, 2 N's, one T and one O. Regard all four S's as one indivisible block. There are $13 - 4 + 1 = 10$ objects to arrange, with repetitions among A, I and N. Thus the answer is $\boxed{10!/(3!2!2!) = 151200}$, option (i).

A(e) Directly, $a_1 = 2(3)+3 = 9$, $a_2 = 2(9)+3 = 21$, $a_3 = 45$, $a_4 = 93$. Thus option $\boxed{\text{(iv)}}$ is correct. For a C.F. + P.I. derivation, write $a_{n+1} - 2a_n = 3$. The C.F. is $C2^n$, and a constant P.I. K gives $K - 2K = 3$, hence $K = -3$. The initial value gives $C - 3 = 3$, so $a_n = 6 \cdot 2^n - 3$.

A(f) Products of even integers are even: $(2r)(2s) = 2(2rs)$. Multiplication is associative, so it is a $\boxed{\text{semigroup}}$, option (ii), the most specific correct choice. It is also closed (a "groupoid" in the elementary algebra sense), but is not a monoid: an identity would satisfy $2e = 2$, forcing $e = 1 \notin 2\mathbb{Z}$.

A(g) The elements of order 3 in S_3 are the two 3-cycles (123) and (132). The identity has order 1, and the three transpositions have order 2. Answer: $\boxed{2}$.

A(h) $K_{m,n}$ is bipartite by its defining two parts; an even cycle is bipartite by alternating its vertices between parts. A complete graph with at least three vertices contains a triangle and is not bipartite. Answer: $\boxed{\text{both (ii) and (iii), option (iv)}}$.

A(i) A tree with at least one edge has chromatic number $\boxed{2}$: choose a root and colour vertices by even/odd distance from it. Each edge joins opposite parities. One colour is impossible because an edge has adjacent endpoints. The one-vertex tree is the exception, with chromatic number 1.

A(j) $p \rightarrow (q \rightarrow p) \equiv \neg p \vee (\neg q \vee p) \equiv (\neg p \vee p) \vee \neg q \equiv \mathrm{T}$. Answer: $\boxed{\text{tautology, option (iii)}}$.

B) Match the following **[10]**

a) Order of element $\bar{3}$ in the group $(Z_7, \times_7)$ is	i) 3
b) Chromatic number of cycle graph C_7 is	ii) 9
c)If $A=\{1,2,3,4,5\}$ then how many elements of A satisfies $(\exists x)(x+1>5)$	iii) 6
d)If any 5 numbers are chosen among 1 to 8 then the sum of any two of them is	iv) 4
e) If |A|=2 and |B|=1 then number of relations from A to B are	v) 1

Solution. B(a) The powers of 3 modulo 7 are $3, 2, 6, 4, 5, 1$. Its order is $\boxed{6}$, match (iii). Strictly, the multiplicative group here must be $\mathbb{Z}_7^*$, not all of $\mathbb{Z}_7$, because 0 has no multiplicative inverse.

B(b) C_7 is an odd cycle. Two alternating colours fail on its closing edge, but a third colour on the last vertex works. Thus $\chi(C_7) = \boxed{3}$, match (i).

B(c) $x + 1 > 5$ means $x > 4$, and the only such member of A is 5. Answer: $\boxed{1}$, match (v). The existential quantifier asserts existence, but the question is asking for the count of satisfying elements.

B(d) Partition $1, \ldots, 8$ into $\{1,8\}, \{2,7\}, \{3,6\}, \{4,5\}$. Choosing five numbers forces some complete pair, whose sum is $\boxed{9}$, match (ii). The wording "any two" is inaccurate: the guarantee is that *at least one pair* has sum 9, not that every pair does.

B(e) A relation is any subset of $A \times B$. That product has $2 \cdot 1 = 2$ elements, and each may independently be included or excluded. Thus there are $2^2 = \boxed{4}$ relations, match (iv).

Question 2

Q. 2) a) Let R be a relation on set $A = \{1,2,3,4\}$. Give examples of R, satisfying the following criterion. Justify your answers. **[6]**

i. R is reflexive and symmetric but not transitive.
ii. R is reflexive and transitive but not symmetric.
iii. R is an symmetric, antisymmetric but not reflexive.

Solution. Let $\Delta = \{(1,1), (2,2), (3,3), (4,4)\}$, the diagonal relation. **(i)** Take $R = \Delta \cup \{(1,2), (2,1), (2,3), (3,2)\}$. All loops are present, so it is reflexive. Every pair has its reverse, so it is symmetric. However $1R2$ and $2R3$ hold while $1R3$ does not, so it is not transitive.

(ii) Take $R = \Delta \cup \{(1,2)\}$. It is reflexive. Any composition of two related pairs either involves a loop (and reproduces a listed pair) or would require a non-loop pair starting at 2, which is absent. Thus it is transitive. It is not symmetric because $(2,1)$ is missing.

(iii) Take $R = \varnothing$. Symmetry and antisymmetry both hold: no pair exists to violate either implication. It is not reflexive because no (a,a) is present. If the scan's awkward wording "an symmetric" was intended to mean *asymmetric*, the same example still works: the empty relation is also asymmetric.

b) In a class, 42% students passed in Mathematics, 45% passed in Physics, 41% passed in Chemistry, 16% passed in Mathematics and Physics, 19% passed in Physics and Chemistry, 18% passed in Chemistry and Mathematics and 15% failed in all three subjects. Find the number of students who passed in all the three subjects if there were 300 students in the class. Also find the number students passed in only one subject. **[6]**

Solution. Work first in percentages. The percentage passing at least one subject is $100 - 15 = 85$. If x percent pass all three, inclusion–exclusion gives

$$85 = 42 + 45 + 41 - 16 - 19 - 18 + x = 75 + x, \quad x = 10.$$

Thus $\boxed{30}$ students pass all three, since 10% of 300 is 30. The only-subject percentages are

$$M : 42 - 16 - 18 + 10 = 18, \quad P : 45 - 16 - 19 + 10 = 20, \quad C : 41 - 18 - 19 + 10 = 14.$$

Therefore 52% pass exactly one, giving $\boxed{156}$ students, with subject counts $54, 60, 42$ respectively.

c) If $f: R-\left\{\frac{7}{3}\right\} \to R-\left\{\frac{4}{3}\right\}$ defined as $f(x)=\frac{4x-5}{3x-7}$. Prove that 'f ' is bijective function, also find the rule for f^{-1}. **[8]**

Solution. The denominator is nonzero on the stated domain. The value $4/3$ is impossible: $(4x-5)/(3x-7) = 4/3$ would imply $12x - 15 = 12x - 28$. **Injective.** If $f(x) = f(z)$, cross-multiplication gives $(4x - 5)(3z - 7) = (4z - 5)(3x - 7)$, hence $-13x = -13z$, so $x = z$. **Surjective and inverse.** For any $y \neq 4/3$, solve

$$y(3x - 7) = 4x - 5, \qquad (3y - 4)x = 7y - 5, \qquad x = \frac{7y - 5}{3y - 4}.$$

The denominator is nonzero, and this input cannot equal $7/3$: that would imply $21y - 15 = 21y - 28$. It therefore belongs to the domain and maps to y. This proves surjectivity and establishes $\boxed{f^{-1}(y) = (7y - 5)/(3y - 4)}$.

Question 3

Q. 3) a) Use mathematical induction to prove that $(11)^{n+2} + (12)^{2n+1}$ is always divisible by 133. **[6]**

Solution. Let $F(n) = 11^{n+2} + 12^{2n+1}$ for nonnegative integers n. At $n = 0$, $F(0) = 121 + 12 = 133$. Assume $F(k) = 133t$. Then

$$\begin{aligned} F(k+1) &= 11 \cdot 11^{k+2} + 144 \cdot 12^{2k+1} \\ &= 11[11^{k+2} + 12^{2k+1}] + (144 - 11)12^{2k+1} \\ &= 133[11t + 12^{2k+1}]. \end{aligned}$$

The bracket is an integer, so the next value is divisible by 133. Induction proves the claim, including all positive integer n.

b) Show that among any 'n+1' numbers one can find two numbers so that their difference is divisible by 'n'. Hence show that among any 12 different two digit numbers one can choose two of them so that their difference is a two digit number with identical first and second digit. **[6]**

Solution. The first statement is about integers: division by n gives only n possible remainders $0, 1, \ldots, n - 1$. Among $n + 1$ integers, two share a remainder. If $a = nq + r$ and $b = ns + r$, then $a - b = n(q - s)$, divisible by n. Apply this with $n = 11$ to the twelve distinct two-digit numbers. Choose the larger as a. Then $a - b$ is a positive multiple of 11 and is at most $99 - 10 = 89$. Thus it is one of $11, 22, 33, 44, 55, 66, 77, 88$, giving the required identical digits. Without the integer assumption, the first statement would not make sense as a divisibility claim.

c) Use Euclidean algorithm to find the greatest common divisor of integers 2002 and 2339 and hence express the GCD in the linear combination of 2002 and 2339. **[8]**

Solution. Repeatedly divide by the previous remainder:

$$\begin{aligned} 2339 &= 1(2002) + 337, \\ 2002 &= 5(337) + 317, \\ 337 &= 1(317) + 20, \\ 317 &= 15(20) + 17, \\ 20 &= 1(17) + 3, \\ 17 &= 5(3) + 2, \\ 3 &= 1(2) + 1, \\ 2 &= 2(1) + 0. \end{aligned}$$

The last nonzero remainder is $\boxed{\gcd(2002, 2339) = 1}$. Back-substitute, preserving the equality at every step:

$$\begin{aligned} 1 &= 3 - 2 = 3 - (17 - 5 \cdot 3) = 6 \cdot 3 - 17 \\ &= 6(20 - 17) - 17 = 6 \cdot 20 - 7 \cdot 17 \\ &= 6 \cdot 20 - 7(317 - 15 \cdot 20) = 111 \cdot 20 - 7 \cdot 317 \\ &= 111(337 - 317) - 7 \cdot 317 = 111 \cdot 337 - 118 \cdot 317 \\ &= 111 \cdot 337 - 118(2002 - 5 \cdot 337) = 701 \cdot 337 - 118 \cdot 2002 \\ &= 701(2339 - 2002) - 118 \cdot 2002 = \boxed{701 \cdot 2339 - 819 \cdot 2002}. \end{aligned}$$

The numerical check is $1{,}639{,}639 - 1{,}639{,}638 = 1$.

Question 4

Q. 4) a) Test the validity of the following argument: It is wrong to refuse to hire the most qualified applicant due to irrelevant criteria. If it is wrong to refuse to hire the most qualified applicant due to irrelevant criteria, then it is wrong to refuse to hire the most qualified applicant due to the color of her skin (because skin color is irrelevant criteria). Therefore, it is wrong to refuse to hire the most qualified applicant due to the color of her skin. **[6]**

Solution. Let p mean it is wrong to refuse the most qualified applicant for irrelevant criteria, and q mean it is wrong to refuse that applicant because of skin colour. The argument has premises p and $p \to q$, with conclusion q. If the premises are true, p is true; the implication can then be true only if q is true. Therefore the argument is valid by modus ponens.

b) Prove that $\sqrt{2}$ is an irrational number. Also prove that $\sqrt{2} + 5$ is an irrational number. **[6]**

Solution. Assume $\sqrt{2} = m/n$ in lowest terms, with $n > 0$. Squaring gives $m^2 = 2n^2$, so m is even, since an odd integer has an odd square. Write $m = 2k$; then $n^2 = 2k^2$, making n even too. This contradicts lowest terms. Hence $\sqrt{2}$ is irrational. If $\sqrt{2}+5$ were rational, subtracting the rational number 5 would make $\sqrt{2}$ rational, which was just disproved. Thus $\boxed{\sqrt{2}+5 \text{ is irrational}}$ as well.

c) Using the **laws** of logic prove the following: **[8]**

$$\sim(p \wedge q) \to \left[\sim p \vee (\sim p \vee q)\right] \equiv \sim p \vee q$$

$$(p \to q) \wedge \left[\sim q \wedge (T \vee \sim q)\right] \leftrightarrow \sim(p \vee q)$$

Solution. First identity:

$$\begin{aligned} \neg(p \wedge q) \to [\neg p \vee (\neg p \vee q)] &\equiv (p \wedge q) \vee \neg p \vee q && \text{implication and idempotence} \\ &\equiv \neg p \vee q. && \text{absorption} \end{aligned}$$

Second identity: Here T denotes the true proposition. Simplify its left-hand side:

$$\begin{aligned} (p \to q) \wedge [\neg q \wedge (T \vee \neg q)] &\equiv (\neg p \vee q) \wedge \neg q && T \vee \neg q \equiv T \\ &\equiv (\neg p \wedge \neg q) \vee (q \wedge \neg q) && \text{distribution} \\ &\equiv \neg p \wedge \neg q \equiv \neg(p \vee q). && \text{De Morgan} \end{aligned}$$

Thus the two sides of the printed biconditional agree on every assignment; that biconditional is a tautology.

Question 5

Q. 5) a) Prove that set of congruence classes Z_6 is a commutative group under addition modulo 6 , '$+_6$' **[6]**

Solution. For $a, b \in \mathbb{Z}_6$, taking the residue of $a + b$ gives an element of $\mathbb{Z}_6$, proving closure. Both bracketings of three modular sums give $a + b + c \pmod 6$, proving associativity. The identity is 0. The inverses of $0, 1, 2, 3, 4, 5$ are respectively $0, 5, 4, 3, 2, 1$, because their sums are 0 modulo 6. Finally $a + b \equiv b + a \pmod 6$, so the operation commutes. Hence it is a commutative (abelian) group.

b) Prove that set of permutations 'S_3' is non-abelian group under composition of functions 'o'. **[6]**

Solution. The elements of S_3 are the six bijections of $\{1, 2, 3\}$: $e, (12), (13), (23), (123), (132)$. The composition of two bijections is a bijection, so the operation is closed. Function composition is associative: for any input x, both $((f \circ g) \circ h)(x)$ and $(f \circ (g \circ h))(x)$ equal $f(g(h(x)))$. The identity permutation fixes all inputs. Each bijection has an inverse bijection; explicitly a transposition is its own inverse, and $(123)^{-1} = (132)$. Therefore S_3 is a group. It is not abelian. With $\alpha = (12)$ and $\beta = (23)$, compose right to left. Then $(\alpha \circ \beta)(1) = 2$, but $(\beta \circ \alpha)(1) = 3$. Thus $\alpha \circ \beta \neq \beta \circ \alpha$.

c) Prove that set of integers $(\mathbb{Z}, \oplus, \otimes)$ is a commutative ring with unity under the operations defined as $a \oplus b = a + b - 1$ and $a \otimes b = a + b - ab$, where $a, b \in \mathbb{Z}$ **[8]**

Solution. Both operation formulas are integer-valued, so they are closed on $\mathbb{Z}$. Addition is associative and commutative:

$$(a \oplus b) \oplus c = a + b + c - 2 = a \oplus (b \oplus c), \quad a \oplus b = b \oplus a.$$

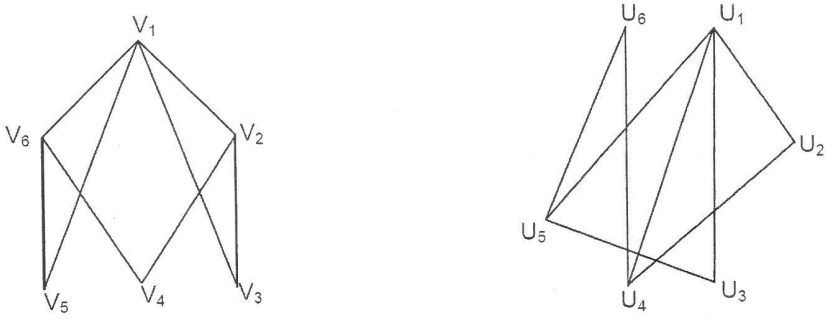
Its identity is 1 and the inverse of a is $2 - a$, because $a \oplus 1 = a$ and $a \oplus (2 - a) = 1$. Thus the additive structure is an abelian group. For multiplication, the two bracketings expand to $a + b + c - ab - ac - bc + abc$; therefore multiplication is associative. The symmetric formula $a + b - ab$ shows it commutes. Distribution holds because

$$a \otimes (b \oplus c) = 2a + b + c - 1 - ab - ac = (a \otimes b) \oplus (a \otimes c),$$

and commutativity supplies the other distributive law. Finally $a \otimes 0 = 0 \otimes a = a$, so the multiplicative unity is $\boxed{0}$. This proves a commutative ring with unity; its additive zero is 1.

Question 6

Q. 6) a) Show that the following graphs are isomorphic. **[6]**



Solution. Define the vertex mapping

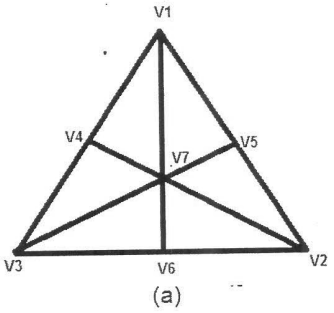
Vertex	v_1	v_2	v_3	v_4	v_5	v_6
Image	u_1	u_4	u_2	u_6	u_3	u_5

It is bijective. To verify adjacency rather than rely on appearance, map the complete edge list:

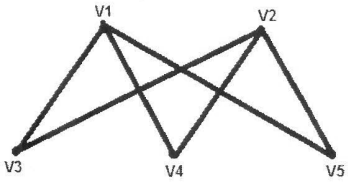
Left edge	Image edge	Left edge	Image edge
v_1v_2	u_1u_4	v_1v_3	u_1u_2
v_1v_5	u_1u_3	v_1v_6	u_1u_5
v_2v_3	u_4u_2	v_2v_4	u_4u_6
v_4v_6	u_6u_5	v_5v_6	u_3u_5

These are exactly the eight edges of the right graph, with none repeated or omitted. Thus the bijection preserves edges and nonedges and is a graph isomorphism.

b) Use the Welch - Powell algorithm to colour the following graphs. Also write down the corresponding chromatic number for each. **[6]**



(a)



(b)

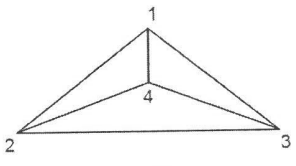
Solution. (a) The central marked vertex v_7 has degree 6; each outer vertex has degree 3. Welsh–Powell with order $v_7, v_1, v_2, v_3, v_4, v_5, v_6$ gives

$$C_1 = \{v_7\}, \quad C_2 = \{v_1, v_2, v_3\}, \quad C_3 = \{v_4, v_5, v_6\}.$$

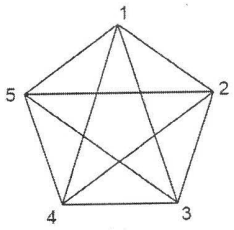
The rim is the even cycle $v_1 - v_4 - v_3 - v_6 - v_2 - v_5 - v_1$, whose two alternating classes are exactly C_2, C_3. A rim edge and the hub form a triangle, so three colours are necessary. Hence $\boxed{\chi = 3}$.

(b) Degrees are $\deg v_1 = \deg v_2 = 3$ and $\deg v_3 = \deg v_4 = \deg v_5 = 2$. The decreasing-degree order is v_1, v_2, v_3, v_4, v_5, giving $C_1 = \{v_1, v_2\}$ and $C_2 = \{v_3, v_4, v_5\}$. The graph is $K_{2,3}$. One colour cannot colour an edge, so $\boxed{\chi = 2}$.

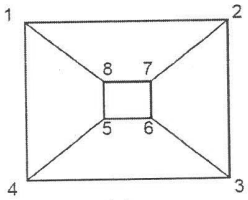
c) Decide which of the following graphs are Eulerian or Hamiltonian or both and write down Eulerian circuit and Hamiltonian circuit wherever exists. **[8]**



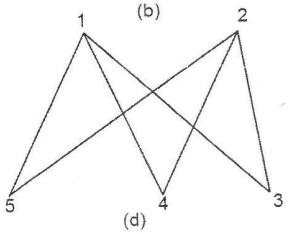
(a)



(b)



(c)



(d)

Solution. (a) K_4**: Hamiltonian, not Eulerian.** Every degree is 3, so no Euler circuit exists. A Hamilton circuit is $\boxed{1-2-3-4-1}$.

(b) K_5**: both.** Every degree is 4, so an Euler circuit exists. One is

$$\boxed{1-2-3-4-5-1-3-5-2-4-1}.$$

Its ten edges are exactly all edges of K_5. A Hamilton circuit is $\boxed{1-2-3-4-5-1}$.

(c) Cube: Hamiltonian, not Eulerian. Each degree is 3, so no Euler circuit exists. A Hamilton circuit following the displayed labels is $\boxed{1-2-3-4-5-6-7-8-1}$.

(d) $K_{2,3}$**: neither.** Vertices $1, 2$ have odd degree 3, so there is no Euler circuit. A cycle in a bipartite graph alternates between its two parts and therefore uses equal numbers from them. A Hamilton cycle would have to use all vertices of parts $\{1, 2\}$ and $\{3, 4, 5\}$, whose sizes differ. It cannot exist. (There is an open Euler trail, but an Euler circuit is what this question asks.)

Question 7

Q. 7) a) If $f : (R,+) \to (R^+, .)$ defined as $f(x) = e^x$, $x \in R$ then show that 'f' is group isomorphism. **[6]**

Solution. For all real x, y, $f(x + y) = e^{x+y} = e^x e^y = f(x)f(y)$, so this is a homomorphism from addition to multiplication. If $e^x = e^y$, taking natural logarithms gives $x = y$, so it is injective. Every positive real t has preimage $x = \ln t$, so it is onto. A bijective homomorphism is an isomorphism, proving the claim, with inverse $f^{-1}(t) = \ln t$.

b) If G = {1, −1, i, −i} is a group under usual complex number multiplication, then prove that H = {1, −1} is a normal subgroup of G, also prove that corresponding G/H is a quotient group. **[6]**

Solution. First H is a subgroup: it is nonempty, products of signs lie in H, its identity is 1, and the inverses of $1, -1$ are themselves. Multiplication of complex numbers commutes, so for any $g \in G$ and $h \in H$, $ghg^{-1} = hgg^{-1} = h \in H$. Thus every conjugate subgroup gHg^{-1} equals H, proving normality. The distinct cosets are $H = \{1, -1\}$ and $iH = \{i, -i\}$. Let $A = H$, $B = iH$. Coset multiplication yields

$\cdot$	A	B
A	A	B
B	B	A

For example $(iH)(iH) = i^2H = (-1)H = H$. The operation is well-defined because H is normal (here changing a representative only multiplies the result by a sign in H). It is associative by multiplication in G; A is the identity; and each coset is its own inverse. Thus G/H is a quotient group, of order 2, isomorphic to $\mathbb{Z}_2$.

c) Use Huffman coding to decode the following fixed length symbols with the frequencies listed: A:37, B:18, C:29, D:13, E:30, F:17, G:6. What is the average number of bits used to encode a character? **[8]**

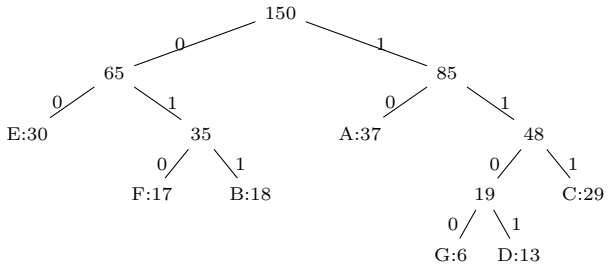
Solution.

Source note. Frequencies alone cannot specify a string to "decode". They specify an alphabet from which to *construct* a Huffman code, as the average-bits question intends. No encoded bit string is supplied.

Huffman repeatedly merges the two smallest current weights:

Step	Merge
1	$G(6) + D(13) = 19$
2	$F(17) + B(18) = 35$
3	$19 + C(29) = 48$
4	$E(30) + 35 = 65$
5	$A(37) + 48 = 85$
6	$65 + 85 = 150$

Label each left branch 0 and right branch 1. The merge tree and one valid code are:



Symbol	Frequency	Code	Length
A	37	10	2
B	18	011	3
C	29	111	3
D	13	1101	4
E	30	00	2
F	17	010	3
G	6	1100	4

No codeword is a prefix of another, since each symbol is a leaf. The mean number of bits is a frequency-weighted average, not an unweighted average of the seven lengths:

$$\bar{L} = \frac{37(2) + 18(3) + 29(3) + 13(4) + 30(2) + 17(3) + 6(4)}{150} = \frac{402}{150} = \boxed{2.68 \text{ bits per character}}.$$

Swapping left and right branches changes codewords but leaves their lengths and this average unchanged.